

TECHNICAL DATABASE AND DATA-STRUCTURE SPECIFICATION

This supplementary document specifies the implemented datasets, SQLite tables, JSON structures, and registered external artifacts that support the methodological protocol. It follows the same Task 1–Task 3 organization. The canonical prototype database is `data/database/corpus_inventory.sqlite`; large text objects, sparse matrices, fitted models, workbooks, and plots remain external files whose paths and hashes are recorded in event or artifact tables.

TASK 1: TEXT PREPARATION AND CORPUS STRUCTURING

Sub-Task 1.1: `document_inventory`

One row per source document; canonical corpus inventory and provenance record.

Variable name	Description	Type / format	Example
<code>document_id</code>	Permanent identifier of the source document.	Text / identifier / primary key	CL_MMA_DEC_000001
<code>document_title</code>	Official title of the source document.	Text	Establece Plan De Prevención Y Descontaminación Atmosférica para la R...
<code>document_type</code>	Recorded document type value for this one row per source document	Text	Decree
<code>document_number</code>	Recorded document number value for this one row per source document	Text	31 Fecha Publicación :24-11-2017 Fecha Promulgación :11-10-2016 Organ...
<code>source_country</code>	Source country retained for lineage.	Text	Chile
<code>source_region</code>	Source region retained for lineage.	Text	Región Metropolitana de Santiago
<code>jurisdiction_scope</code>	Recorded jurisdiction scope value for this one row per source document	Text	Regional plan issued by national ministry; applies to Región Metropol...
<code>issuing_institution</code>	Recorded issuing institution value for this one row per source document	Text	MINISTERIO DEL MEDIO AMBIENTE Título :ESTABLECE PLAN DE PREVENCIÓN Y ...
<code>publication_date</code>	Date associated with publication, in ISO format.	Text / ISO date	2017-11-24
<code>promulgation_date</code>	Date associated with promulgation, in ISO format.	Text / ISO date	2016-10-11
<code>effective_date</code>	Date associated with effective, in ISO format.	Text / ISO date	2017-11-24
<code>version_type</code>	Recorded version type value for this one row per source document	Text	Única
<code>source_url_repository</code>	Source url repository retained for lineage.	Text	https://www.leychile.cl/N?i=1111283&f=2017-11-24&p= ESTABLECE PLAN DE...
<code>source_database</code>	Source database retained for lineage.	Text	LeyChile / Biblioteca del Congreso Nacional de Chile
<code>source_external_id</code>	Identifier of source external.	Text / identifier	1111283 URL :https://www.leychile.cl/N?i=1111283&f=2017-11-24&p= ESTA...
<code>language</code>	Recorded language value for this one row per source document	Text	Spanish (es)
<code>original_file_format</code>	Recorded original file format value for this one row per source document	Text	Searchable PDF
<code>file_name_original</code>	Recorded file name original value for this one row per source document	Text	DTO-31_24-Establece-plan-de-prevencion-y-descontaminacion-atmosferica...
<code>file_size_bytes</code>	Recorded file size bytes value for this one row per source document	Integer	2020239
<code>page_count</code>	Number of page.	Integer	90
<code>file_hash_sha256</code>	SHA-256 integrity hash of file.	Text / 64-character SHA-256	5d1293a34ead...

Variable name	Description	Type / format	Example
repository_generated_date	Date associated with repository generated, in ISO format.	Text / ISO date	2018-05-22
access_date	Date associated with access, in ISO format.	Text / ISO date	2026-07-10
ingestion_date	Date associated with ingestion, in ISO format.	Text / ISO date	2026-07-10
inventory_created_at	Recorded inventory created at value for this one row per source document	Text	2026-07-10
document_status	Completion or review status for document.	Text	Ready for processing
notes	Optional free-text warnings, limitations, or implementation notes.	Text	Initial inspection found text on 90 of 90 pages. Tables/figures shoul...

Sub-Task 1.2: extraction_events

One row per extraction attempt; links a document to raw TXT and page-level JSON.

Implemented foreign-key links: document_id → document_inventory.document_id.

Variable name	Description	Type / format	Example
extraction_id	Identifier of the extraction event.	Text / identifier / primary key	EXT_000001
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
extraction_method	Method used for extraction. Required in the implemented schema.	Text	PDF text extraction
extraction_tool	Software or script used for extraction. Required in the implemented schema.	Text	pdfplumber
tool_version	Version identifier for tool.	Text	0.11.10
extraction_date	Date associated with extraction, in ISO format. Required in the implemented schema.	Text / ISO date	2026-07-10
ocr_language	Recorded ocr language value for this one row per extraction attempt	Text	Not applicable
page_count	Number of page.	Integer	90
extracted_page_count	Number of extracted page.	Integer	90
layout_preserved	Recorded layout preserved value for this one row per extraction attempt	Text	Partially
raw_text_path	Project-relative path to raw text.	Text / project-relative path	data/raw_text/CL_MMA_DEC_000001/EXT_000001_raw.txt
page_json_path	Project-relative path to page json.	Text / project-relative path	data/raw_text/CL_MMA_DEC_000001/EXT_000001_pages.json
extraction_status	Completion or review status for extraction. Required in the implemented schema.	Text	Completed
extraction_notes	Recorded extraction notes value for this one row per extraction attempt	Text	null
source_file_hash_sha256	SHA-256 integrity hash of source file.	Text / 64-character SHA-256	5d1293a34ead...
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-07-10T21:51:42+00:00

Sub-Task 1.3: quality_assessment_events

One row per document-level extraction-quality assessment.

Implemented foreign-key links: document_id → document_inventory.document_id; extraction_id → extraction_events.extraction_id.

Variable name	Description	Type / format	Example
assessment_id	Identifier of the extraction-quality assessment.	Text / identifier / primary key	QA_000001
extraction_id	Identifier of the extraction event. Required in the implemented schema.	Text / identifier	EXT_000001
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
assessment_date	Date associated with assessment, in ISO format. Required in the implemented schema.	Text / ISO date	2026-07-10
quality_tool	Software or script used for quality. Required in the implemented schema.	Text	03_assess_extraction_quality.py
near_blank_threshold	Recorded near blank threshold value for this one row per document-level extraction-quality assessment. Required in the implemented schema.	Integer	50
expected_page_count	Number of expected page.	Integer	90
observed_page_count	Number of observed page. Required in the implemented schema.	Integer	90
page_count_match	Recorded page count match value for this one row per document-level extraction-quality assessment. Required in the implemented schema.	Integer / Boolean (0 or 1)	1
blank_page_count	Number of blank page. Required in the implemented schema.	Integer	0

Variable name	Description	Type / format		Example
near_blank_page_count	Number of near blank page. Required in the implemented schema.	Integer	0	
replacement_char_page_count	Number of replacement char page. Required in the implemented schema.	Integer	0	
manual_review_page_count	Number of manual review page. Required in the implemented schema.	Integer	0	
min_nonspace_chars	Recorded min nonspace chars value for this one row per document-level extraction-quality assessment.	Integer	83	
max_nonspace_chars	Recorded max nonspace chars value for this one row per document-level extraction-quality assessment.	Integer	4570	
mean_nonspace_chars	Recorded mean nonspace chars value for this one row per document-level extraction-quality assessment.	Real number	1800.7300	
quality_status	Completion or review status for quality. Required in the implemented schema.	Text	Completed	
quality_checked_json_path	Project-relative path to quality checked json. Required in the implemented schema.	Text / project-relative path	data/quality_checked/CL_MMA_DEC_000001/EXT_000001_quality_checked.json	
page_quality_csv_path	Project-relative path to page quality csv. Required in the implemented schema.	Text / project-relative path	tables/quality_reports/CL_MMA_DEC_000001/EXT_000001_page_quality.csv	
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-07-10T22:41:03+00:00	

Variable name	Description	Type / format	Example
notes	Optional free-text warnings, limitations, or implementation notes.	Text null	
review_reason_summary	Recorded review reason summary value for this one row per document-level extraction-quality assessment.	Text null	

Sub-Task 1.3: quality_page_flags

One row per assessed page and quality-assessment event.

Implemented foreign-key links: `document_id` → `document_inventory.document_id`; `extraction_id` → `extraction_events.extraction_id`; `assessment_id` → `quality_assessment_events.assessment_id`.

Variable name	Description	Type / format	Example
assessment_id	Identifier of the extraction-quality assessment. Required in the implemented schema.	Text / identifier / primary key	QA_000001
extraction_id	Identifier of the extraction event. Required in the implemented schema.	Text / identifier	EXT_000001
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
page_number	Recorded page number value for this one row per assessed page and quality-assessment event. Required in the implemented schema.	Integer / composite primary key part 2	1
total_char_count	Number of total char. Required in the implemented schema.	Integer	4073
nonspace_char_count	Number of nonspace char. Required in the implemented schema.	Integer	3466
word_count	Number of word. Required in the implemented schema.	Integer	642
line_count	Number of line. Required in the implemented schema.	Integer	62
blank_extraction	Recorded blank extraction value for this one row per assessed page and quality-assessment event. Required in the implemented schema.	Integer / Boolean (0 or 1)	0
near_blank_extraction	Recorded near blank extraction value for this one row per assessed page and quality-assessment event. Required in the implemented schema.	Integer / Boolean (0 or 1)	0
replacement_char_count	Number of replacement char. Required in the implemented schema.	Integer	0
requires_manual_review	Indicates whether human review is required (0 = no; 1 = yes). Required in the implemented schema.	Integer / Boolean (0 or 1)	0
quality_status	Completion or review status for quality. Required in the implemented schema.	Text	text_extracted
notes	Optional free-text warnings, limitations, or implementation notes.	Text	null

Variable name	Description	Type / format	Example
review_reason_code	Machine-readable reason for manual review. Required in the implemented schema. Default: 'none'.	Text	none
review_reason_note	Human-readable explanation of the review flag.	Text	null

Sub-Task 1.4: normalization_events

One row per normalization run and normalized JSON output.

Implemented foreign-key links: document_id → document_inventory.document_id; extraction_id → extraction_events.extraction_id; assessment_id → quality_assessment_events.assessment_id.

Variable name	Description	Type / format	Example
normalization_id	Identifier of the normalization event.	Text / identifier / primary key	NORM_000001
assessment_id	Identifier of the extraction-quality assessment. Required in the implemented schema.	Text / identifier	QA_000001
extraction_id	Identifier of the extraction event. Required in the implemented schema.	Text / identifier	EXT_000001
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
normalization_date	Date associated with normalization, in ISO format. Required in the implemented schema.	Text / ISO date	2026-07-10
normalization_tool	Software or script used for normalization. Required in the implemented schema.	Text	04_normalize_text.py
normalization_rules_json	JSON object describing the normalization or matching-only transformations applied. Required in the implemented schema.	Text / serialized JSON	{"unicode_normalization":"NFC composition normalization for main norm...
pages_normalized_count	Number of pages normalized. Required in the implemented schema.	Integer	90
pages_with_warnings_count	Number of pages with warnings. Required in the implemented schema.	Integer	0
normalized_json_path	Project-relative path to normalized json. Required in the implemented schema.	Text / project-relative path	data/normalized/CL_MMA_DEC_000001/QA_000001_NORM_000001_normalized.json
page_normalization_csv_path	Project-relative path to page normalization csv. Required in the implemented schema.	Text / project-relative path	tables/normalization_reports/CL_MMA_DEC_000001/QA_000001_NORM_000001_...
normalization_status	Completion or review status for normalization. Required in the implemented schema.	Text	Completed
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-07-10T23:52:54+00:00
notes	Optional free-text warnings, limitations, or implementation notes.	Text	null

Sub-Task 1.4: normalization_page_stats

One row per normalized page and normalization event.

Implemented foreign-key links: document_id → document_inventory.document_id; normalization_id → normalization_events.normalization_id.

Variable name	Description	Type / format	Example
normalization_id	Identifier of the normalization event. Required in the implemented schema.	Text / identifier / primary key	NORM_000001
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001

Variable name	Description	Type / format	Example
page_number	Recorded page number value for this one row per normalized page and normalization event. Required in the implemented schema.	Integer / composite primary key part 2	1
raw_char_count	Number of raw char. Required in the implemented schema.	Integer	4073
normalized_char_count	Number of normalized char. Required in the implemented schema.	Integer	3977
matching_text_char_count	Number of matching text char. Required in the implemented schema.	Integer	3977
removed_technical_lines_count	Number of removed technical lines. Required in the implemented schema.	Integer	1
replacement_char_count	Number of replacement char. Required in the implemented schema.	Integer	0
quality_requires_manual_review	Recorded quality requires manual review value for this one row per normalized page and normalization event. Required in the implemented schema.	Integer	0
normalization_status	Completion or review status for normalization. Required in the implemented schema.	Text	completed
normalization_warning	Recorded normalization warning value for this one row per normalized page and normalization event.	Text	null

Sub-Task 1.5: `structure_events`

One row per legal-structure parser execution.

Implemented foreign-key links: `normalization_id` → `normalization_events.normalization_id`.

Variable name	Description	Type / format	Example
structure_id	Identifier of the legal-structure event.	Text / identifier / primary key	STRUCT_000001
normalization_id	Identifier of the normalization event. Required in the implemented schema.	Text / identifier	NORM_000001
assessment_id	Identifier of the extraction-quality assessment. Required in the implemented schema.	Text / identifier	QA_000001
extraction_id	Identifier of the extraction event. Required in the implemented schema.	Text / identifier	EXT_000001
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
structure_date	Date associated with structure, in ISO format. Required in the implemented schema.	Text / ISO date	2026-07-14
structure_tool	Software or script used for structure. Required in the implemented schema.	Text	05_identify_legal_structure.py
parser_method	Method used for parser. Required in the implemented schema.	Text	regular_expression_state_parser
structure_version	Version identifier for structure. Required in the implemented schema.	Text	0.1
source_text_field	Source text field retained for lineage. Required in the implemented schema.	Text	normalization.normalized_text
elements_count	Number of elements. Required in the implemented schema.	Integer	1245
uncertain_elements_count	Number of uncertain elements. Required in the implemented schema.	Integer	1
manual_review_elements_count	Number of manual review elements. Required in the implemented schema.	Integer	0
structure_json_path	Project-relative path to structure json. Required in the implemented schema.	Text / project-relative path	data/legal_structure/CL_MMA_DEC_000001/NORM_000001_STRUCT_000001_stru...
structure_csv_path	Project-relative path to structure csv. Required in the implemented schema.	Text / project-relative path	tables/legal_structure/CL_MMA_DEC_000001/NORM_000001_STRUCT_000001_st...
structure_status	Completion or review status for structure. Required in the implemented schema.	Text	Completed with warnings
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-07-14T16:57:52+00:00

Variable name	Description	Type / format	Example
notes	Optional free-text warnings, limitations, or implementation notes.	Text	1 element(s) have uncertain structure.

Sub-Task 1.5: structure_elements

One row per flattened node in the parsed legal hierarchy.

Implemented foreign-key links: structure_id → structure_events.structure_id.

Variable name	Description	Type / format	Example
structure_id	Identifier of the legal-structure event. Required in the implemented schema.	Text / identifier / primary key	STRUCT_000001
element_id	Stable identifier of a legal-structure element. Required in the implemented schema.	Text / identifier / composite primary key part 2	CL_MMA_DEC_000001_META_001
parent_id	Identifier of the element's immediate parent; null for root nodes.	Text / identifier	ID_000001
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
element_order	Sequential source order of the structural element. Required in the implemented schema.	Integer	1
nesting_level	Depth of the node in the legal hierarchy; root nodes use level 0. Required in the implemented schema.	Integer	0
element_type	Recorded element type value for this one row per flattened node in the parsed legal hierarchy. Required in the implemented schema.	Text	metadata_block; metadata_field; other; preamble; paragraph; chapter; article; graph
label	Recorded label value for this one row per flattened node in the parsed legal hierarchy.	Text	null
title	Recorded title value for this one row per flattened node in the parsed legal hierarchy.	Text	null
page_start	Recorded page start value for this one row per flattened node in the parsed legal hierarchy. Required in the implemented schema.	Integer	1
page_end	Recorded page end value for this one row per flattened node in the parsed legal hierarchy. Required in the implemented schema.	Integer	1
element_text	Recorded element text value for this one row per flattened node in the parsed legal hierarchy.	Text	null
text_source	Recorded text source value for this one row per flattened node in the parsed legal hierarchy.	Text	null
content_status	Completion or review status for content.	Text	text_extracted; title or caption extracted
structure_confidence	Recorded structure confidence value for this one row per flattened node in the parsed legal hierarchy. Required in the implemented schema.	Text	high; uncertain; medium
requires_manual_review	Indicates whether human review is required (0 = no; 1 = yes). Required in the implemented schema.	Integer / Boolean (0 or 1)	0
review_reason_code	Machine-readable reason for manual review.	Text	none
review_reason_note	Human-readable explanation of the review flag.	Text	null

Task 1 external datasets and file roles

Sub - task	Artifact	Forma t	Role / path pattern
1.1	Document inventory export	CSV and JSON	tables/inventory/document_inventory.*; human-readable and exchange representations of document_inventory.
1.2	Raw page-marked text	TXT	data/raw_text/<document_id>/<extraction_id>_raw.txt; immutable extracted text with explicit page markers.
1.2	Page extraction record	JSON	data/raw_text/<document_id>/<extraction_id>_pages.json; raw text and extraction metadata by page.
1.3	Quality-checked pages	JSON	data/quality_checked/<document_id>/<extraction_id>_quality_checked.json.
1.3	Page-quality report	CSV	tables/quality_reports/<document_id>/<extraction_id>_page_quality.csv.
1.4	Normalized pages	JSON	data/normalized/<document_id>/<assessment_id>_<normalization_id>_normalized.json.
1.4	Page-normalization report	CSV	tables/normalization_reports/<document_id>/<assessment_id>_<normalization_id>_page_normalization.csv.
1.5	Hierarchical legal structure	JSON	data/legal_structure/<document_id>/<normalization_id>_<structure_id>_structure.json.
1.5	Flattened legal structure	CSV	tables/legal_structure/<document_id>/<normalization_id>_<structure_id>_structure_elements.csv.

Sub-Task 1.1 Document-inventory JSON

Array of document_inventory-compatible records used as an exchange export.

```
[
  {
    "document_id": "CL_MMA_DEC_000001",
    "document_title": "Establece Plan De Prevención Y Descontaminación Atmosférica para la R...",
    "document_type": "Decree",
    "document_number": "31 Fecha Publicación :24-11-2017 Fecha Promulgación :11-10-2016 Organ...",
    "source_country": "Chile",
    "source_region": "Región Metropolitana de Santiago",
    "jurisdiction_scope": "Regional plan issued by national ministry; applies to Región Metropol...",
    "issuing_institution": "MINISTERIO DEL MEDIO AMBIENTE Título :ESTABLECE PLAN DE PREVENCIÓN Y ...",
    "publication_date": "2017-11-24",
    "promulgation_date": "2016-10-11",
    "effective_date": "2017-11-24",
    "version_type": "Única",
    "source_url_repository": "https://www.leychile.cl/N?i=1111283&f=2017-11-24&p=ESTABLECE PLAN DE...",
    "source_database": "LeyChile / Biblioteca del Congreso Nacional de Chile",
    "source_external_id": "1111283 URL :https://www.leychile.cl/N?i=1111283&f=2017-11-24&p= ESTA...",
    "language": "Spanish (es)",
    "original_file_format": "Searchable PDF",
    "file_name_original": "DTO-31_24-Establece-plan-de-prevencion-y-descontaminacion-atmosferica...",
    "file_size_bytes": "2020239",
    "page_count": "90",
  }
]
```

```

    "file_hash_sha256":
"5d1293a34ead7f3702473109b64b9f8f87237b6c674d4f35b3a6f647921c9fdd",
    "repository_generated_date": "2018-05-22",
    "access_date": "2026-07-10",
    "ingestion_date": "2026-07-10",
    "inventory_created_at": "2026-07-10",
    "document_status": "Ready for processing",
    "notes": "Initial inspection found text on 90 of 90 pages. Tables/figures
shoul..."
  }
]

```

Sub-Task 1.2 Page-level extraction JSON

Preserves document and extraction metadata and the unmodified raw text for each PDF page.

```

{
  "document_id": "CL_MMA_DEC_000001",
  "document_metadata": {
    "document_title": "Establece Plan De Prevención Y Descontaminación Atmosférica
para la R...",
    "language": "Spanish (es)",
    "original_file_format": "Searchable PDF",
    "file_name_original": "DTO-31_24-Establece-plan-de-prevencion-y-
descontaminacion-atmosferica..."
  },
  "extraction_metadata": {
    "extraction_id": "EXT_000001",
    "extraction_method": "PDF text extraction",
    "extraction_tool": "pdfplumber",
    "tool_version": "0.11.10",
    "extraction_date": "2026-07-10T21:51:42+00:00",
    "ocr_language": "null",
    "source_pdf_path": "documents/source_pdfs/DTO-31_24-Establece-plan-de-
prevencion-y-descon...",
    "source_file_hash_sha256":
"5d1293a34ead7f3702473109b64b9f8f87237b6c674d4f35b3a6f647921c9fdd",
    "source_page_count": "90",
    "layout_preserved": "Partially",
    "raw_text_path": "data/raw_text/CL_MMA_DEC_000001/EXT_000001_raw.txt"
  },
  "pages": [
    "{...}"
  ]
}

```

JSON path	Description	Type / format	Example
\$.document_id	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
\$.document_metadata	Recorded document metadata value for this record	object	{...}
\$.document_metadata.document_title	Official title of the source document.	str	Establece Plan De Prevención Y Descontaminación Atmosférica para la R...
\$.document_metadata.language	Recorded language value for this record	str	Spanish (es)
\$.document_metadata.original_file_format	Recorded original file format value for this record	str	Searchable PDF
\$.document_metadata.file_name_original	Recorded file name original value for this record	str	DTO-31_24-Establece-plan-de-prevencion-y-descontaminacion-atmosferica...
\$.extraction_metadata	Recorded extraction metadata value for this record	object	{...}

JSON path	Description	Type / format	Example
<code>\$.extraction_metadata.extraction_id</code>	Identifier of the extraction event.	str	EXT_000001
<code>\$.extraction_metadata.extraction_method</code>	Method used for extraction.	str	PDF text extraction
<code>\$.extraction_metadata.extraction_tool</code>	Software or script used for extraction.	str	pdfplumber
<code>\$.extraction_metadata.tool_version</code>	Version identifier for tool.	str	0.11.10
<code>\$.extraction_metadata.extraction_date</code>	Date associated with extraction, in ISO format.	str	2026-07-10T21:51:42+00:00
<code>\$.extraction_metadata.ocr_language</code>	Recorded ocr language value for this record	NoneType	null
<code>\$.extraction_metadata.source_pdf_path</code>	Project-relative path to source pdf.	str	documents/source_pdfs/DTO-31_24-Establece-plan-de-prevencion-y-descon...
<code>\$.extraction_metadata.source_file_hash_sha256</code>	SHA-256 integrity hash of source file.	str	5d1293a34ead...
<code>\$.extraction_metadata.source_page_count</code>	Number of source page.	int	90
<code>\$.extraction_metadata.layout_preserved</code>	Recorded layout preserved value for this record	str	Partially
<code>\$.extraction_metadata.raw_text_path</code>	Project-relative path to raw text.	str	data/raw_text/CL_MMA_DEC_000001/EXT_000001_raw.txt
<code>\$.pages</code>	Recorded pages value for this record	array	[...]
<code>\$.pages[]</code>	One item in the pages array.	object	{...}
<code>\$.pages[].page_number</code>	Recorded page number value for this record	int	1
<code>\$.pages[].raw_text</code>	Recorded raw text value for this record	str	Tipo Norma :Decreto 31 Fecha Publicación :24-11-2017 Fecha Promulgaci...
<code>\$.pages[].extraction_method</code>	Method used for extraction.	str	PDF text extraction

Sub-Task 1.3 Quality-checked JSON

Extends the page-level extraction record with assessment metadata, page flags, and a document summary.

```
{
  "document_id": "CL_MMA_DEC_000001",
  "document_metadata": {
    "document_title": "Establece Plan De Prevención Y Descontaminación Atmosférica para la R...",
    "language": "Spanish (es)",
    "original_file_format": "Searchable PDF",
    "file_name_original": "DTO-31_24-Establece-plan-de-prevencion-y-descontaminacion-atmosferica..."
  },
  "extraction_metadata": {
    "extraction_id": "EXT_000001",
    "extraction_method": "PDF text extraction",
    "extraction_tool": "pdfplumber",
    "tool_version": "0.11.10",
    "extraction_date": "2026-07-10T21:51:42+00:00",
    "ocr_language": "null",
    "source_pdf_path": "documents/source_pdfs/DTO-31_24-Establece-plan-de-prevencion-y-descon...",
    "source_file_hash_sha256": "5d1293a34ead7f3702473109b64b9f8f87237b6c674d4f35b3a6f647921c9fdd",
    "source_page_count": "90",
    "layout_preserved": "Partially",
    "raw_text_path": "data/raw_text/CL_MMA_DEC_000001/EXT_000001_raw.txt"
  },
  "pages": [
    {...}
  ],
  "quality_assessment_metadata": {
    "assessment_id": "QA_000002",
    "extraction_id": "EXT_000001",

```

```

    "document_id": "CL_MMA_DEC_000001",
    "assessment_date": "2026-07-10T22:41:03+00:00",
    "quality_tool": "03_assess_extraction_quality.py",
    "near_blank_threshold": "50"
  },
  "document_level_quality_summary": {
    "expected_page_count": "90",
    "observed_page_count": "90",
    "page_count_match": "True",
    "blank_page_count": "0",
    "near_blank_page_count": "0",
    "replacement_char_page_count": "0",
    "manual_review_page_count": "0",
    "review_reason_summary": "null",
    "min_nonspace_chars": "83",
    "max_nonspace_chars": "4570",
    "mean_nonspace_chars": "1800.7300",
    "near_blank_threshold": "50",
    "quality_status": "Completed",
    "notes": "null"
  }
}

```

JSON path	Description	Type / format	Example
\$.document_id	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
\$.document_metadata	Recorded document metadata value for this record	object	{...}
\$.document_metadata.document_title	Official title of the source document.	str	Establece Plan De Prevención Y Descontaminación Atmosférica para la R...
\$.document_metadata.language	Recorded language value for this record	str	Spanish (es)
\$.document_metadata.original_file_format	Recorded original file format value for this record	str	Searchable PDF
\$.document_metadata.file_name_original	Recorded file name original value for this record	str	DTO-31_24-Establece-plan-de-prevencion-y-descontaminacion-atmosferica...
\$.extraction_metadata	Recorded extraction metadata value for this record	object	{...}
\$.extraction_metadata.extraction_id	Identifier of the extraction event.	str	EXT_000001
\$.extraction_metadata.extraction_method	Method used for extraction.	str	PDF text extraction
\$.extraction_metadata.extraction_tool	Software or script used for extraction.	str	pdfplumber
\$.extraction_metadata.tool_version	Version identifier for tool.	str	0.11.10
\$.extraction_metadata.extraction_date	Date associated with extraction, in ISO format.	str	2026-07-10T21:51:42+00:00
\$.extraction_metadata.ocr_language	Recorded ocr language value for this record	NoneType	null
\$.extraction_metadata.source_pdf_path	Project-relative path to source pdf.	str	documents/source_pdfs/DTO-31_24-Establece-plan-de-prevencion-y-descon...
\$.extraction_metadata.source_file_hash_sha256	SHA-256 integrity hash of source file.	str	5d1293a34ead...
\$.extraction_metadata.source_page_count	Number of source page.	int	90
\$.extraction_metadata.layout_preserved	Recorded layout preserved value for this record	str	Partially
\$.extraction_metadata.raw_text_path	Project-relative path to raw text.	str	data/raw_text/CL_MMA_DEC_000001/EXT_000001_raw.txt
\$.pages	Recorded pages value for this record	array	[...]
\$.pages[]	One item in the pages array.	object	{...}
\$.pages[].page_number	Recorded page number value for this record	int	1
\$.pages[].raw_text	Recorded raw text value for this record	str	Tipo Norma :Decreto 31 Fecha Publicación :24-11-2017 Fecha Promulgaci...
\$.pages[].extraction_method	Method used for extraction.	str	PDF text extraction

JSON path	Description	Type / format	Example
<code>\$.pages[].quality_assessment</code>	Recorded quality assessment value for this record	object	{...}
<code>\$.pages[].quality_assessment.page_number</code>	Recorded page number value for this record	int	1
<code>\$.pages[].quality_assessment.total_char_count</code>	Number of total char.	int	4073
<code>\$.pages[].quality_assessment.nonspace_char_count</code>	Number of nonspace char.	int	3466
<code>\$.pages[].quality_assessment.word_count</code>	Number of word.	int	642
<code>\$.pages[].quality_assessment.line_count</code>	Number of line.	int	62
<code>\$.pages[].quality_assessment.blank_extraction</code>	Recorded blank extraction value for this record	bool	False
<code>\$.pages[].quality_assessment.near_blank_extraction</code>	Recorded near blank extraction value for this record	bool	False
<code>\$.pages[].quality_assessment.replacement_char_count</code>	Number of replacement char.	int	0
<code>\$.pages[].quality_assessment.requires_manual_review</code>	Indicates whether human review is required (0 = no; 1 = yes).	bool	False
<code>\$.pages[].quality_assessment.review_reason_code</code>	Machine-readable reason for manual review.	str	none
<code>\$.pages[].quality_assessment.review_reason_note</code>	Human-readable explanation of the review flag.	NoneType	null
<code>\$.pages[].quality_assessment.quality_status</code>	Completion or review status for quality.	str	text_extracted
<code>\$.pages[].quality_assessment.notes</code>	Optional free-text warnings, limitations, or implementation notes.	NoneType	null
<code>\$.quality_assessment_metadata</code>	Recorded quality assessment metadata value for this record	object	{...}
<code>\$.quality_assessment_metadata.assessment_id</code>	Identifier of the extraction-quality assessment.	str	QA_000002
<code>\$.quality_assessment_metadata.extraction_id</code>	Identifier of the extraction event.	str	EXT_000001
<code>\$.quality_assessment_metadata.document_id</code>	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
<code>\$.quality_assessment_metadata.assessment_date</code>	Date associated with assessment, in ISO format.	str	2026-07-10T22:41:03+00:00
<code>\$.quality_assessment_metadata.quality_tool</code>	Software or script used for quality.	str	03_assess_extraction_quality.py
<code>\$.quality_assessment_metadata.near_blank_threshold</code>	Recorded near blank threshold value for this record	int	50
<code>\$.document_level_quality_summary</code>	Recorded document level quality summary value for this record	object	{...}
<code>\$.document_level_quality_summary.expected_page_count</code>	Number of expected page.	int	90
<code>\$.document_level_quality_summary.observed_page_count</code>	Number of observed page.	int	90
<code>\$.document_level_quality_summary.page_count_match</code>	Recorded page count match value for this record	bool	True
<code>\$.document_level_quality_summary.blank_page_count</code>	Number of blank page.	int	0
<code>\$.document_level_quality_summary.near_blank_page_count</code>	Number of near blank page.	int	0
<code>\$.document_level_quality_summary.replacement_char_page_count</code>	Number of replacement char page.	int	0
<code>\$.document_level_quality_summary.manual_review_page_count</code>	Number of manual review page.	int	0
<code>\$.document_level_quality_summary.review_reason_summary</code>	Recorded review reason summary value for this record	NoneType	null
<code>\$.document_level_quality_summary.min_nonspace_chars</code>	Recorded min nonspace chars value for this record	int	83
<code>\$.document_level_quality_summary.max_nonspace_chars</code>	Recorded max nonspace chars value for this record	int	4570
<code>\$.document_level_quality_summary.mean_nonspace_chars</code>	Recorded mean nonspace chars value for this record	float	1800.7300
<code>\$.document_level_quality_summary.near_blank_threshold</code>	Recorded near blank threshold value for this record	int	50
<code>\$.document_level_quality_summary.quality_status</code>	Completion or review status for quality.	str	Completed

JSON path	Description	Type / format	Example
\$.document_level_quality_summaries	Optional free-text warnings, limitations, or implementation notes.	NoneType	null

Sub-Task 1.4 Normalized page-level JSON

Retains raw text and quality fields while adding normalized_text, matching_text, normalization steps, warnings, and summaries.

```
{
  "document_id": "CL_MMA_DEC_000001",
  "document_metadata": {
    "document_title": "Establece Plan De Prevención Y Descontaminación Atmosférica para la R...",
    "language": "Spanish (es)",
    "original_file_format": "Searchable PDF",
    "file_name_original": "DTO-31_24-Establece-plan-de-prevencion-y-descontaminacion-atmosferica..."
  },
  "extraction_metadata": {
    "extraction_id": "EXT_000001",
    "extraction_method": "PDF text extraction",
    "extraction_tool": "pdfplumber",
    "tool_version": "0.11.10",
    "extraction_date": "2026-07-10T21:51:42+00:00",
    "ocr_language": "null",
    "source_pdf_path": "documents/source_pdfs/DTO-31_24-Establece-plan-de-prevencion-y-descon...",
    "source_file_hash_sha256": "5d1293a34ead7f3702473109b64b9f8f87237b6c674d4f35b3a6f647921c9fdd",
    "source_page_count": "90",
    "layout_preserved": "Partially",
    "raw_text_path": "data/raw_text/CL_MMA_DEC_000001/EXT_000001_raw.txt"
  },
  "pages": [
    "{...}"
  ],
  "quality_assessment_metadata": {
    "assessment_id": "QA_000002",
    "extraction_id": "EXT_000001",
    "document_id": "CL_MMA_DEC_000001",
    "assessment_date": "2026-07-10T22:41:03+00:00",
    "quality_tool": "03_assess_extraction_quality.py",
    "near_blank_threshold": "50"
  },
  "document_level_quality_summary": {
    "expected_page_count": "90",
    "observed_page_count": "90",
    "page_count_match": "True",
    "blank_page_count": "0",
    "near_blank_page_count": "0",
    "replacement_char_page_count": "0",
    "manual_review_page_count": "0",
    "review_reason_summary": "null",
    "min_nonspace_chars": "83",
    "max_nonspace_chars": "4570",
    "mean_nonspace_chars": "1800.7300",
    "near_blank_threshold": "50",
    "quality_status": "Completed",
    "notes": "null"
  },
  "normalization_metadata": {
```

```

    "normalization_id": "NORM_000004",
    "assessment_id": "QA_000001",
    "extraction_id": "EXT_000001",
    "document_id": "CL_MMA_DEC_000001",
    "normalization_date": "2026-07-23T20:02:32+00:00",
    "normalization_tool": "04_normalize_text.py",
    "normalization_rules": "{...}"
  },
  "document_level_normalization_summary": {
    "normalization_id": "NORM_000004",
    "assessment_id": "QA_000001",
    "extraction_id": "EXT_000001",
    "document_id": "CL_MMA_DEC_000001",
    "pages_normalized_count": "90",
    "pages_with_warnings_count": "0",
    "normalization_status": "Completed",
    "notes": "null"
  }
}

```

JSON path	Description	Type / format	Example
\$.document_id	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
\$.document_metadata	Recorded document metadata value for this record	object	{...}
\$.document_metadata.document_title	Official title of the source document.	str	Establece Plan De Prevención Y Descontaminación Atmosférica para la R...
\$.document_metadata.language	Recorded language value for this record	str	Spanish (es)
\$.document_metadata.original_file_format	Recorded original file format value for this record	str	Searchable PDF
\$.document_metadata.file_name_original	Recorded file name original value for this record	str	DTO-31_24-Establece-plan-de-prevencion-y-descontaminacion-atmosferica...
\$.extraction_metadata	Recorded extraction metadata value for this record	object	{...}
\$.extraction_metadata.extraction_id	Identifier of the extraction event.	str	EXT_000001
\$.extraction_metadata.extraction_method	Method used for extraction.	str	PDF text extraction
\$.extraction_metadata.extraction_tool	Software or script used for extraction.	str	pdfplumber
\$.extraction_metadata.tool_version	Version identifier for tool.	str	0.11.10
\$.extraction_metadata.extraction_date	Date associated with extraction, in ISO format.	str	2026-07-10T21:51:42+00:00
\$.extraction_metadata.ocr_language	Recorded ocr language value for this record	NoneType	null
\$.extraction_metadata.source_pdf_path	Project-relative path to source pdf.	str	documents/source_pdfs/DTO-31_24-Establece-plan-de-prevencion-y-descon...
\$.extraction_metadata.source_file_hash_sha256	SHA-256 integrity hash of source file.	str	5d1293a34ead...
\$.extraction_metadata.source_page_count	Number of source page.	int	90
\$.extraction_metadata.layout_preserved	Recorded layout preserved value for this record	str	Partially
\$.extraction_metadata.raw_text_path	Project-relative path to raw text.	str	data/raw_text/CL_MMA_DEC_000001/EXT_000001_raw.txt
\$.pages	Recorded pages value for this record	array	[...]
\$.pages[]	One item in the pages array.	object	{...}
\$.pages[].page_number	Recorded page number value for this record	int	1
\$.pages[].raw_text	Recorded raw text value for this record	str	Tipo Norma :Decreto 31 Fecha Publicación :24-11-2017 Fecha Promulgaci...
\$.pages[].extraction_method	Method used for extraction.	str	PDF text extraction
\$.pages[].quality_assessment	Recorded quality assessment value for this record	object	{...}
\$.pages[].quality_assessment.page_number	Recorded page number value for this record	int	1
\$.pages[].quality_assessment.total_char_count	Number of total char.	int	4073

JSON path	Description	Type / format	Example
<code>\$.pages[].quality_assessment.nonspace_char_count</code>	Number of nonspace char.	int	3466
<code>\$.pages[].quality_assessment.word_count</code>	Number of word.	int	642
<code>\$.pages[].quality_assessment.line_count</code>	Number of line.	int	62
<code>\$.pages[].quality_assessment.blank_extraction</code>	Recorded blank extraction value for this record	bool	False
<code>\$.pages[].quality_assessment.near_blank_extraction</code>	Recorded near blank extraction value for this record	bool	False
<code>\$.pages[].quality_assessment.replacement_char_count</code>	Number of replacement char.	int	0
<code>\$.pages[].quality_assessment.requires_manual_review</code>	Indicates whether human review is required (0 = no; 1 = yes).	bool	False
<code>\$.pages[].quality_assessment.review_reason_code</code>	Machine-readable reason for manual review.	str	none
<code>\$.pages[].quality_assessment.review_reason_note</code>	Human-readable explanation of the review flag.	NoneType	null
<code>\$.pages[].quality_assessment.quality_statuses</code>	Completion or review status for quality.	str	text_extracted
<code>\$.pages[].quality_assessment.notes</code>	Optional free-text warnings, limitations, or implementation notes.	NoneType	null
<code>\$.pages[].normalization</code>	Recorded normalization value for this record	object	{...}
<code>\$.pages[].normalization.normalized_text</code>	Recorded normalized text value for this record	str	Tipo Norma:Decreto 31 Fecha Publicación:24-11-2017 Fecha Promulgación...
<code>\$.pages[].normalization.matching_text</code>	Recorded matching text value for this record	str	tipo norma:decreto 31 fecha publicacion:24-11-2017 fecha promulgacion...
<code>\$.pages[].normalization.normalization_steps</code>	Recorded normalization steps value for this record	array	[...]
<code>\$.pages[].normalization.normalization_steps[]</code>	One item in the normalization_steps array.	str	unicode_normalization
<code>\$.pages[].normalization.normalization_status</code>	Completion or review status for normalization.	str	completed
<code>\$.pages[].normalization.normalization_warning</code>	Recorded normalization warning value for this record	NoneType	null
<code>\$.quality_assessment_metadata</code>	Recorded quality assessment metadata value for this record	object	{...}
<code>\$.quality_assessment_metadata.assessment_id</code>	Identifier of the extraction-quality assessment.	str	QA_000002
<code>\$.quality_assessment_metadata.extraction_id</code>	Identifier of the extraction event.	str	EXT_000001
<code>\$.quality_assessment_metadata.document_id</code>	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
<code>\$.quality_assessment_metadata.assessment_date</code>	Date associated with assessment, in ISO format.	str	2026-07-10T22:41:03+00:00
<code>\$.quality_assessment_metadata.quality_tool</code>	Software or script used for quality.	str	03_assess_extraction_quality.py
<code>\$.quality_assessment_metadata.near_blank_threshold</code>	Recorded near blank threshold value for this record	int	50
<code>\$.document_level_quality_summary</code>	Recorded document level quality summary value for this record	object	{...}
<code>\$.document_level_quality_summary.expected_page_count</code>	Number of expected page.	int	90
<code>\$.document_level_quality_summary.observed_page_count</code>	Number of observed page.	int	90
<code>\$.document_level_quality_summary.page_count_match</code>	Recorded page count match value for this record	bool	True
<code>\$.document_level_quality_summary.blank_page_count</code>	Number of blank page.	int	0
<code>\$.document_level_quality_summary.near_blank_page_count</code>	Number of near blank page.	int	0
<code>\$.document_level_quality_summary.replacement_char_page_count</code>	Number of replacement char page.	int	0
<code>\$.document_level_quality_summary.manual_review_page_count</code>	Number of manual review page.	int	0
<code>\$.document_level_quality_summary.review_reason_summary</code>	Recorded review reason summary value for this record	NoneType	null
<code>\$.document_level_quality_summary.min_nonspace_chars</code>	Recorded min nonspace chars value for this record	int	83

JSON path	Description	Type / format	Example
<code>\$.document_level_quality_summary.max_nonspace_chars</code>	Recorded max nonspace chars value for this record	int	4570
<code>\$.document_level_quality_summary.mean_nonspace_chars</code>	Recorded mean nonspace chars value for this record	float	1800.7300
<code>\$.document_level_quality_summary.near_blank_threshold</code>	Recorded near blank threshold value for this record	int	50
<code>\$.document_level_quality_summary.quality_status</code>	Completion or review status for quality.	str	Completed
<code>\$.document_level_quality_summary.notes</code>	Optional free-text warnings, limitations, or implementation notes.	NoneType	null
<code>\$.normalization_metadata</code>	Recorded normalization metadata value for this record	object	{...}
<code>\$.normalization_metadata.normalization_id</code>	Identifier of the normalization event.	str	NORM_000004
<code>\$.normalization_metadata.assessment_id</code>	Identifier of the extraction-quality assessment.	str	QA_000001
<code>\$.normalization_metadata.extraction_id</code>	Identifier of the extraction event.	str	EXT_000001
<code>\$.normalization_metadata.document_id</code>	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
<code>\$.normalization_metadata.normalization_date</code>	Date associated with normalization, in ISO format.	str	2026-07-23T20:02:32+00:00
<code>\$.normalization_metadata.normalization_tool</code>	Software or script used for normalization.	str	04_normalize_text.py
<code>\$.normalization_metadata.normalization_rules</code>	Recorded normalization rules value for this record	object	{...}
<code>\$.normalization_metadata.normalization_rules.unicode_normalization</code>	Recorded unicode normalization value for this record	str	NFC composition normalization for main normalized text
<code>\$.normalization_metadata.normalization_rules.targeted_compatibility_replacements</code>	Recorded targeted compatibility replacements value for this record	str	typographic ligatures and selected symbols only; legal N°/Nº markers ...
<code>\$.normalization_metadata.normalization_rules.quotation_mark_standardization</code>	Recorded quotation mark standardization value for this record	str	typographic quotes converted to plain quotes
<code>\$.normalization_metadata.normalization_rules.dash_standardization</code>	Recorded dash standardization value for this record	str	en/em/minus dashes converted to hyphen-minus
<code>\$.normalization_metadata.normalization_rules.space_normalization</code>	Recorded space normalization value for this record	str	non-breaking and irregular spaces converted to ordinary spaces
<code>\$.normalization_metadata.normalization_rules.soft_hyphen_removal</code>	Recorded soft hyphen removal value for this record	str	soft hyphen and line-break hyphenation removed conservatively
<code>\$.normalization_metadata.normalization_rules.zero_width_character_removal</code>	Recorded zero width character removal value for this record	str	zero-width characters removed
<code>\$.normalization_metadata.normalization_rules.technical_content_removal</code>	Recorded technical content removal value for this record	str	explicit repository headers/footers and HTML tags removed from normal...
<code>\$.normalization_metadata.normalization_rules.legal_structure_preservation</code>	Recorded legal structure preservation value for this record	str	legal structure markers are preserved as block-start markers; title/c...
<code>\$.normalization_metadata.normalization_rules.metadata_line_preservation</code>	Recorded metadata line preservation value for this record	str	repository metadata key-value lines are preserved as separate lines
<code>\$.normalization_metadata.normalization_rules.matching_text</code>	Recorded matching text value for this record	str	accent-insensitive lowercase auxiliary field generated separately fro...
<code>\$.document_level_normalization_summary</code>	Recorded document level normalization summary value for this record	object	{...}
<code>\$.document_level_normalization_summary.normalization_id</code>	Identifier of the normalization event.	str	NORM_000004
<code>\$.document_level_normalization_summary.assessment_id</code>	Identifier of the extraction-quality assessment.	str	QA_000001
<code>\$.document_level_normalization_summary.extraction_id</code>	Identifier of the extraction event.	str	EXT_000001
<code>\$.document_level_normalization_summary.document_id</code>	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
<code>\$.document_level_normalization_summary.pages_normalized_count</code>	Number of pages normalized.	int	90
<code>\$.document_level_normalization_summary.pages_with_warnings_count</code>	Number of pages with warnings.	int	0
<code>\$.document_level_normalization_summary.normalization_status</code>	Completion or review status for normalization.	str	Completed
<code>\$.document_level_normalization_summary.notes</code>	Optional free-text warnings, limitations, or implementation notes.	NoneType	null

1.5 Hierarchical legal-structure JSON

Represents the legal document as nested structural nodes. The same node schema is recursively reused under children[].

```
{
  "document_id": "CL_MMA_DEC_000001",
  "document_metadata": {
    "document_title": "Establece Plan De Prevención Y Descontaminación Atmosférica
para la R...",
    "document_type": "Decree",
    "source_country": "Chile",
    "source_region": "Región Metropolitana de Santiago",
    "issuing_institution": "MINISTERIO DEL MEDIO AMBIENTE Título :ESTABLECE PLAN DE
PREVENCIÓN Y ...",
    "publication_date": "2017-11-24",
    "promulgation_date": "2016-10-11",
    "effective_date": "2017-11-24",
    "source_url": "https://www.leychile.cl/N?i=1111283&f=2017-11-24&p= ESTABLECE
PLAN DE...",
    "language": "Spanish (es)",
    "original_file_format": "Searchable PDF"
  },
  "workflow_metadata": {
    "structure_id": "STRUCT_000010",
    "normalization_id": "NORM_000004",
    "assessment_id": "QA_000001",
    "extraction_id": "EXT_000001",
    "structure_created_by": "rule_based_parser",
    "structure_tool": "05_identify_legal_structure.py",
    "parser_method": "regular_expression_state_parser",
    "structure_version": "0.6",
    "source_text_field": "normalization.normalized_text"
  },
  "source_page_range": {
    "start": "1",
    "end": "90"
  },
  "structure_summary": {
    "elements_count": "907",
    "uncertain_elements_count": "0",
    "manual_review_elements_count": "0",
    "structure_status": "Completed",
    "notes": "null"
  },
  "structure": [
    "{...}"
  ]
}
```

JSON path	Description	Type / format	Example
\$.document_id	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
\$.document_metadata	Recorded document metadata value for this record	object	{...}
\$.document_metadata.document_title	Official title of the source document.	str	Establece Plan De Prevención Y Descontaminación Atmosférica para la R...
\$.document_metadata.document_type	Recorded document type value for this record	str	Decree
\$.document_metadata.source_country	Source country retained for lineage.	str	Chile
\$.document_metadata.source_region	Source region retained for lineage.	str	Región Metropolitana de Santiago

JSON path	Description	Type / format	Example
<code>\$.document_metadata.issuing_institution</code>	Recorded issuing institution value for this record	str	MINISTERIO DEL MEDIO AMBIENTE Título :ESTABLECE PLAN DE PREVENCIÓN Y ...
<code>\$.document_metadata.publication_date</code>	Date associated with publication, in ISO format.	str	2017-11-24
<code>\$.document_metadata.promulgation_date</code>	Date associated with promulgation, in ISO format.	str	2016-10-11
<code>\$.document_metadata.effective_date</code>	Date associated with effective, in ISO format.	str	2017-11-24
<code>\$.document_metadata.source_url</code>	URL for source.	str	https://www.leychile.cl/N?i=1111283&f=2017-11-24&p= ESTABLECE PLAN DE...
<code>\$.document_metadata.language</code>	Recorded language value for this record	str	Spanish (es)
<code>\$.document_metadata.original_file_format</code>	Recorded original file format value for this record	str	Searchable PDF
<code>\$.workflow_metadata</code>	Recorded workflow metadata value for this record	object	{...}
<code>\$.workflow_metadata.structure_id</code>	Identifier of the legal-structure event.	str	STRUCT_000010
<code>\$.workflow_metadata.normalization_id</code>	Identifier of the normalization event.	str	NORM_000004
<code>\$.workflow_metadata.assessment_id</code>	Identifier of the extraction-quality assessment.	str	QA_000001
<code>\$.workflow_metadata.extraction_id</code>	Identifier of the extraction event.	str	EXT_000001
<code>\$.workflow_metadata.structure_created_by</code>	Recorded structure created by value for this record	str	rule_based_parser
<code>\$.workflow_metadata.structure_tool</code>	Software or script used for structure.	str	05_identify_legal_structure.py
<code>\$.workflow_metadata.parser_method</code>	Method used for parser.	str	regular_expression_state_parser
<code>\$.workflow_metadata.structure_version</code>	Version identifier for structure.	str	0.6
<code>\$.workflow_metadata.source_text_field</code>	Source text field retained for lineage.	str	normalization.normalized_text
<code>\$.source_page_range</code>	Source page range retained for lineage.	object	{...}
<code>\$.source_page_range.start</code>	Recorded start value for this record	int	1
<code>\$.source_page_range.end</code>	Recorded end value for this record	int	90
<code>\$.structure_summary</code>	Recorded structure summary value for this record	object	{...}
<code>\$.structure_summary.elements_count</code>	Number of elements.	int	907
<code>\$.structure_summary.uncertain_elements_count</code>	Number of uncertain elements.	int	0
<code>\$.structure_summary.manual_review_elements_count</code>	Number of manual review elements.	int	0
<code>\$.structure_summary.structure_status</code>	Completion or review status for structure.	str	Completed
<code>\$.structure_summary.notes</code>	Optional free-text warnings, limitations, or implementation notes.	NoneType	null
<code>\$.structure</code>	Recorded structure value for this record	array	[...]
<code>\$.structure[]</code>	One item in the structure array.	object	{...}

JSON path	Description	Type / format	Example
\$.structure[].element_id	Stable identifier of a legal-structure element.	str	CL_MMA_DEC_000001_META_001
\$.structure[].parent_id	Identifier of the element's immediate parent; null for root nodes.	NoneType	ID_000001
\$.structure[].element_type	Recorded element type value for this record	str	metadata_block
\$.structure[].label	Recorded label value for this record	NoneType	null
\$.structure[].title	Recorded title value for this record	NoneType	null
\$.structure[].page_start	Recorded page start value for this record	int	1
\$.structure[].page_end	Recorded page end value for this record	int	1
\$.structure[].structure_confidence	Recorded structure confidence value for this record	str	high
\$.structure[].requires_manual_review	Indicates whether human review is required (0 = no; 1 = yes).	bool	False
\$.structure[].review_reason_code	Machine-readable reason for manual review.	str	none
\$.structure[].review_reason_note	Human-readable explanation of the review flag.	NoneType	null
\$.structure[].children	Recorded children value for this record	array	[...]
\$.structure[].children[]	One item in the children array.	object	{...}
\$.structure[].children[].element_id	Stable identifier of a legal-structure element.	str	CL_MMA_DEC_000001_META_001_FIELD_001
\$.structure[].children[].parent_id	Identifier of the element's immediate parent; null for root nodes.	str	CL_MMA_DEC_000001_META_001
\$.structure[].children[].element_type	Recorded element type value for this record	str	metadata_field
\$.structure[].children[].label	Recorded label value for this record	str	Tipo Norma
\$.structure[].children[].title	Recorded title value for this record	NoneType	null
\$.structure[].children[].page_start	Recorded page start value for this record	int	1
\$.structure[].children[].page_end	Recorded page end value for this record	int	1
\$.structure[].children[].structure_confidence	Recorded structure confidence value for this record	str	high
\$.structure[].children[].requires_manual_review	Indicates whether human review is required (0 = no; 1 = yes).	bool	False
\$.structure[].children[].review_reason_code	Machine-readable reason for manual review.	str	none
\$.structure[].children[].review_reason_note	Human-readable explanation of the review flag.	NoneType	null
\$.structure[].children[].children	Recorded children value for this record	array	[...]
\$.structure[].children[].element_text	Recorded element text value for this record	str	Tipo Norma :Decreto 31

JSON path	Description	Type / format	Example
\$.structure[].children[].text_source	Recorded text source value for this record	str	normalization.normalized_text
\$.structure[].children[].content_status	Completion or review status for content.	str	text_extracted

Sub-Task 1.3 Page-quality CSV

Variable name	Description	Type / format	Example
page_number	Recorded page number value for this record	Text	1
total_char_count	Number of total char.	Text	4073
nonspace_char_count	Number of nonspace char.	Text	3466
word_count	Number of word.	Text	642
line_count	Number of line.	Text	62
blank_extraction	Recorded blank extraction value for this record	Boolean	False
near_blank_extraction	Recorded near blank extraction value for this record	Boolean	False
replacement_char_count	Number of replacement char.	Text	0
requires_manual_review	Indicates whether human review is required (0 = no; 1 = yes).	Boolean	False
review_reason_code	Machine-readable reason for manual review.	Text	none
review_reason_note	Human-readable explanation of the review flag.	Text	null
quality_status	Completion or review status for quality.	Text	text_extracted
notes	Optional free-text warnings, limitations, or implementation notes.	Text	null

Sub-Task 1.4 Page-normalization CSV

Variable name	Description	Type / format	Example
page_number	Recorded page number value for this record	Text	1
raw_char_count	Number of raw char.	Text	4073
normalized_char_count	Number of normalized char.	Text	3967
matching_text_char_count	Number of matching text char.	Text	3967
removed_technical_lines_count	Number of removed technical lines.	Text	1
replacement_char_count	Number of replacement char.	Text	0
quality_requires_manual_review	Recorded quality requires manual review value for this record	Boolean	False
normalization_status	Completion or review status for normalization.	Text	completed
normalization_warning	Recorded normalization warning value for this record	Text	null

Sub-Task 1.5 Flattened structure CSV

Variable name	Description	Type / format	Example
structure_id	Identifier of the legal-structure event.	Text	STRUCT_000010
element_id	Stable identifier of a legal-structure element.	Text	CL_MMA_DEC_000001_META_001
parent_id	Identifier of the element's immediate parent; null for root nodes.	Text	ID_000001
document_id	Permanent identifier of the source document.	Text	CL_MMA_DEC_000001
element_order	Sequential source order of the structural element.	Integer	1

Variable name	Description	Type / format	Example
nesting_level	Depth of the node in the legal hierarchy; root nodes use level 0.	Text	0
element_type	Recorded element type value for this record	Text	metadata_block
label	Recorded label value for this record	Text	null
title	Recorded title value for this record	Text	null
page_start	Recorded page start value for this record	Text	1
page_end	Recorded page end value for this record	Text	1
element_text	Recorded element text value for this record	Text	null
text_source	Recorded text source value for this record	Text	null
content_status	Completion or review status for content.	Text	null
structure_confidence	Recorded structure confidence value for this record	Text	high
requires_manual_review	Indicates whether human review is required (0 = no; 1 = yes).	Text	0
review_reason_code	Machine-readable reason for manual review.	Text	none
review_reason_note	Human-readable explanation of the review flag.	Text	null

TASK 2: SEGMENTATION INTO CANDIDATE STATEMENTS

Task 2: segmentation_events

One row per candidate-statement segmentation run.

Implemented foreign-key links: document_id → document_inventory.document_id; structure_id → structure_events.structure_id.

Variable name	Description	Type / format	Example
segmentation_id	Identifier of the candidate-statement segmentation event.	Text / identifier / primary key	SEG_000001
structure_id	Identifier of the legal-structure event. Required in the implemented schema.	Text / identifier	STRUCT_000009
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
segmentation_date	Date associated with segmentation, in ISO format. Required in the implemented schema.	Text / ISO date	2026-07-23

Variable name	Description	Type / format	Example
segmentation_tool	Software or script used for segmentation. Required in the implemented schema.	Text	06_segment_candidate_statements.py
spacy_model	Recorded spacy model value for this one row per candidate-statement segmentation run. Required in the implemented schema.	Text	es_core_news_md
spacy_version	Version identifier for spacy. Required in the implemented schema.	Text	3.8.14
segmentation_version	Version identifier for segmentation. Required in the implemented schema.	Text	0.1
source_elements_count	Number of source elements. Required in the implemented schema.	Integer	608
candidate_statements_count	Number of candidate statements. Required in the implemented schema.	Integer	773
manual_review_count	Number of manual review. Required in the implemented schema.	Integer	159
candidate_json_path	Project-relative path to candidate json. Required in the implemented schema.	Text / project-relative path	data/statement_segmentation/CL_MMA_DEC_000001/STRUCT_000009_SEG_00000
candidate_csv_path	Project-relative path to candidate csv. Required in the implemented schema.	Text / project-relative path	tables/statement_segmentation/CL_MMA_DEC_000001/STRUCT_000009_SEG_000

Variable name	Description	Type / format	Example
segmentation_status	Completion or review status for segmentation. Required in the implemented schema.	Text	Completed with warnings
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-07-23T15:53:28+00:00
notes	Optional free-text warnings, limitations, or implementation notes.	Text	missing_terminal_punctuation: 128; sentence_starts_lowercase: 31; ver...

Task 2: candidate_statements

One row per sentence-level candidate statement.

Implemented foreign-key links: `structure_id` → `structure_elements.structure_id`; `source_element_id` → `structure_elements.element_id`; `document_id` → `document_inventory.document_id`; `segmentation_id` → `segmentation_events.segmentation_id`.

Variable name	Description	Type / format	Example
statement_id	Stable identifier of the candidate statement. Required in the implemented schema.	Text / identifier / composite primary key part 2	CL_MMA_DEC_000001_STMT_000001
segmentation_id	Identifier of the candidate-statement segmentation event. Required in the implemented schema.	Text / identifier / primary key	SEG_000001
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
structure_id	Identifier of the legal-structure event. Required in the implemented schema.	Text / identifier	STRUCT_000009
source_element_id	Identifier of the legal-structure element from which a candidate was segmented. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001_PREAMBLE_VISTO_P001
source_parent_id	Identifier of the source element's parent.	Text / identifier	CL_MMA_DEC_000001_PREAMBLE_VISTO

Variable name	Description	Type / format		Example
source_element_type	Source element type retained for lineage. Required in the implemented schema.	Text	paragraph	
source_element_order	Source element order retained for lineage. Required in the implemented schema.	Integer	12	
sentence_order_within_element	Sentence position within the source structural element. Required in the implemented schema.	Integer	1	
statement_order	Global sequential order of the candidate in the document. Required in the implemented schema.	Integer	1	
statement_text	Original candidate-statement text; never overwritten by transformed or predicted text. Required in the implemented schema.	Text	Lo dispuesto en la ley N°19.300, sobre Bases Generales del Medio Ambi...	
source_char_start	Inclusive starting character offset in the source element text. Required in the implemented schema.	Integer	0	
source_char_end	Exclusive ending character offset in the source element text. Required in the implemented schema.	Integer	2015	
page_start	Recorded page start value for this one row per sentence-level candidate statement. Required in the implemented schema.	Integer	1	
page_end	Recorded page end value for this one row per sentence-level candidate statement. Required in the implemented schema.	Integer	1	
page_assignment_method	Method used for page assignment. Required in the implemented schema.	Text	element_inherited	
segmentation_method	Method used for segmentation. Required in the implemented schema.	Text	spacy_sentence_boundaries_with_legal_postprocessing	
segmentation_status	Completion or review status for segmentation. Required in the implemented schema.	Text	completed_with_warnings; completed	

Variable name	Description	Type / format	Example
requires_manual_review	Indicates whether human review is required (0 = no; 1 = yes). Required in the implemented schema.	Integer / Boolean (0 or 1)	1
review_reason_code	Machine-readable reason for manual review.	Text	missing_terminal_punctuation; very_short_sentence; sentence_starts_lowercase; very_short_sentence;sentence_starts_lowercase;missing_terminal_punctu...; very_short_sentence;missing_terminal_punctuation
review_reason_note	Human-readable explanation of the review flag.	Text	Candidate lacks final period, question mark, or exclamation mark; lis...
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-07-23T15:53:28+00:00

Task 2 external datasets and JSON

Artifact	Format	Role / path pattern
Candidate-statement JSON	JSON	data/statement_segmentation/<document_id>/<structure_id>_<segmentation_id>_candidate_statements.json; full metadata, warnings, merges, and candidates.
Candidate-statement table	CSV	tables/statement_segmentation/<document_id>/<structure_id>_<segmentation_id>_candidate_statements.csv; one row per candidate.

2. Candidate-statement JSON

Stores document and structure lineage, segmentation metadata, technical quality checks, warning summaries, protected-boundary merges, and candidate records.

```
{
  "document_id": "CL_MMA_DEC_000001",
  "document_metadata": {
    "document_title": "Establece Plan De Prevención Y Descontaminación Atmosférica para la R...",
    "document_type": "Decree",
    "source_country": "Chile",
    "source_region": "Región Metropolitana de Santiago",
    "issuing_institution": "MINISTERIO DEL MEDIO AMBIENTE Título :ESTABLECE PLAN DE PREVENCIÓN Y ...",
    "publication_date": "2017-11-24",
    "promulgation_date": "2016-10-11",
    "effective_date": "2017-11-24",
    "source_url": "https://www.leychile.cl/N?i=1111283&f=2017-11-24&p= ESTABLECE PLAN DE...",
    "language": "Spanish (es)",
    "original_file_format": "Searchable PDF"
  },
  "structure_event_metadata": {
    "structure_id": "STRUCT_000010",
    "normalization_id": "NORM_000004",
    "assessment_id": "QA_000001",
    "extraction_id": "EXT_000001",
    "document_id": "CL_MMA_DEC_000001",
    "structure_date": "2026-07-23",
    "structure_tool": "05_identify_legal_structure.py",
    "parser_method": "regular_expression_state_parser",
    "structure_version": "0.6",
  }
}
```

```

        "source_text_field": "normalization.normalized_text",
        "elements_count": "907",
        "uncertain_elements_count": "0",
        "manual_review_elements_count": "0",
        "structure_json_path":
"data/legal_structure/CL_MMA_DEC_000001/NORM_000004_STRUCT_000010_stru...",
        "structure_csv_path":
"tables/legal_structure/CL_MMA_DEC_000001/NORM_000004_STRUCT_000010_st...",
        "structure_status": "Completed",
        "created_at": "2026-07-23T20:05:13+00:00",
        "notes": "null"
    },
    "segmentation_metadata": {
        "segmentation_id": "SEG_000003",
        "structure_id": "STRUCT_000010",
        "document_id": "CL_MMA_DEC_000001",
        "segmentation_date": "2026-07-23",
        "segmentation_tool": "06_segment_candidate_statements.py",
        "spacy_model": "es_core_news_md",
        "spacy_version": "3.8.14",
        "segmentation_version": "0.2",
        "source_elements_count": "599",
        "candidate_statements_count": "767",
        "manual_review_count": "144",
        "candidate_json_path":
"data/statement_segmentation/CL_MMA_DEC_000001/STRUCT_000010_SEG_00000...",
        "candidate_csv_path":
"tables/statement_segmentation/CL_MMA_DEC_000001/STRUCT_000010_SEG_000...",
        "segmentation_status": "Completed with warnings",
        "created_at": "2026-07-23T20:21:02+00:00",
        "notes": "missing_terminal_punctuation: 119; sentence_starts_lowercase: 24;
ver...",
        "included_element_types": "[...]",
        "page_assignment_method": "element_inherited",
        "segmentation_method": "spacy_sentence_boundaries_with_legal_postprocessing",
        "source_char_end_semantics": "exclusive",
        "merged_protected_boundaries_count": "19"
    },
    "quality_checks": {
        "no_empty_candidate_statements": "True",
        "candidate_statement_order_is_sequential": "True",
        "every_statement_has_valid_source_element_id": "True",
        "every_nonempty_processed_element_produces_statement": "True",
        "character_offsets_reproduce_statement_text": "True",
        "protected_legal_abbreviation_boundary_violations": "0",
        "protected_boundary_violation_details": "[...]"
    },
    "warning_summary": {
        "missing_terminal_punctuation": "119",
        "sentence_starts_lowercase": "24",
        "very_short_sentence": "8"
    },
    "merged_protected_boundaries": [
        "{...}"
    ],
    "candidate_statements": [
        "{...}"
    ]
}

```

JSON path	Description	Type / format	Example
\$.document_id	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
\$.document_metadata	Recorded document metadata value for this record	object	{...}
\$.document_metadata.document_title	Official title of the source document.	str	Establece Plan De Prevención Y Descontaminación Atmosférica para la R...
\$.document_metadata.document_type	Recorded document type value for this record	str	Decree
\$.document_metadata.source_country	Source country retained for lineage.	str	Chile
\$.document_metadata.source_region	Source region retained for lineage.	str	Región Metropolitana de Santiago
\$.document_metadata.issuing_institution	Recorded issuing institution value for this record	str	MINISTERIO DEL MEDIO AMBIENTE Título :ESTABLECE PLAN DE PREVENCIÓN Y ...
\$.document_metadata.publication_date	Date associated with publication, in ISO format.	str	2017-11-24
\$.document_metadata.promulgation_date	Date associated with promulgation, in ISO format.	str	2016-10-11
\$.document_metadata.effective_date	Date associated with effective, in ISO format.	str	2017-11-24
\$.document_metadata.source_url	URL for source.	str	https://www.leychile.cl/N?i=1111283&f=2017-11-24&p=ESTABLECE PLAN DE...
\$.document_metadata.language	Recorded language value for this record	str	Spanish (es)
\$.document_metadata.original_file_format	Recorded original file format value for this record	str	Searchable PDF
\$.structure_event_metadata	Recorded structure event metadata value for this record	object	{...}
\$.structure_event_metadata.structure_id	Identifier of the legal-structure event.	str	STRUCT_000010
\$.structure_event_metadata.normalization_id	Identifier of the normalization event.	str	NORM_000004
\$.structure_event_metadata.assessment_id	Identifier of the extraction-quality assessment.	str	QA_000001
\$.structure_event_metadata.extraction_id	Identifier of the extraction event.	str	EXT_000001
\$.structure_event_metadata.document_id	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
\$.structure_event_metadata.structure_date	Date associated with structure, in ISO format.	str	2026-07-23
\$.structure_event_metadata.structure_tool	Software or script used for structure.	str	05_identify_legal_structure.py
\$.structure_event_metadata.parser_method	Method used for parser.	str	regular_expression_state_parser
\$.structure_event_metadata.structure_version	Version identifier for structure.	str	0.6
\$.structure_event_metadata.source_text_field	Source text field retained for lineage.	str	normalization.normalized_text
\$.structure_event_metadata.elements_count	Number of elements.	int	907
\$.structure_event_metadata.uncertain_elements_count	Number of uncertain elements.	int	0
\$.structure_event_metadata.manual_review_elements_count	Number of manual review elements.	int	0

JSON path	Description	Type / format		Example
<code>\$.structure_event_metadata.structure_json_path</code>	Project-relative path to structure json.	str		data/legal_structure/CL_MMA_DEC_000001/NORM_000004_STRUCT_000010_stru...
<code>\$.structure_event_metadata.structure_csv_path</code>	Project-relative path to structure csv.	str		tables/legal_structure/CL_MMA_DEC_000001/NORM_000004_STRUCT_000010_st...
<code>\$.structure_event_metadata.structure_status</code>	Completion or review status for structure.	str		Completed
<code>\$.structure_event_metadata.created_at</code>	Timestamp at which the record was created, in ISO 8601 format.	str		2026-07-23T20:05:13+00:00
<code>\$.structure_event_metadata.notes</code>	Optional free-text warnings, limitations, or implementation notes.	NoneType	null	
<code>\$.segmentation_metadata</code>	Recorded segmentation metadata value for this record	object		{...}
<code>\$.segmentation_metadata.segmentation_id</code>	Identifier of the candidate-statement segmentation event.	str		SEG_000003
<code>\$.segmentation_metadata.structure_id</code>	Identifier of the legal-structure event.	str		STRUCT_000010
<code>\$.segmentation_metadata.document_id</code>	Permanent identifier of the source document.	str		CL_MMA_DEC_000001
<code>\$.segmentation_metadata.segmentation_date</code>	Date associated with segmentation, in ISO format.	str		2026-07-23
<code>\$.segmentation_metadata.segmentation_tool</code>	Software or script used for segmentation.	str		06_segment_candidate_statements.py
<code>\$.segmentation_metadata.spacy_model</code>	Recorded spacy model value for this record	str		es_core_news_md
<code>\$.segmentation_metadata.spacy_version</code>	Version identifier for spacy.	str		3.8.14
<code>\$.segmentation_metadata.segmentation_version</code>	Version identifier for segmentation.	str		0.2
<code>\$.segmentation_metadata.source_elements_count</code>	Number of source elements.	int		599
<code>\$.segmentation_metadata.candidate_statements_count</code>	Number of candidate statements.	int		767
<code>\$.segmentation_metadata.manual_review_count</code>	Number of manual review.	int		144
<code>\$.segmentation_metadata.candidate_json_path</code>	Project-relative path to candidate json.	str		data/statement_segmentation/CL_MMA_DEC_000001/STRUCT_000010_SEG_00000...
<code>\$.segmentation_metadata.candidate_csv_path</code>	Project-relative path to candidate csv.	str		tables/statement_segmentation/CL_MMA_DEC_000001/STRUCT_000010_SEG_000...
<code>\$.segmentation_metadata.segmentation_status</code>	Completion or review status for segmentation.	str		Completed with warnings
<code>\$.segmentation_metadata.created_at</code>	Timestamp at which the record was created, in ISO 8601 format.	str		2026-07-23T20:21:02+00:00
<code>\$.segmentation_metadata.notes</code>	Optional free-text warnings, limitations, or implementation notes.	str		missing_terminal_punctuation: 119; sentence_starts_lowercase: 24; ver...

JSON path	Description	Type / format		Example
\$.segmentation_metadata.included_element_types	Recorded included element types value for this record	array	[...]	
\$.segmentation_metadata.included_element_types[]	One item in the included_element_types array.	str	paragraph	
\$.segmentation_metadata.page_assignment_method	Method used for page assignment.	str	element_inherited	
\$.segmentation_metadata.segmentation_method	Method used for segmentation.	str	spacy_sentence_boundaries_with_legal_postprocessing	
\$.segmentation_metadata.source_char_end_semantics	Source char end semantics retained for lineage.	str	exclusive	
\$.segmentation_metadata.merged_protected_boundaries_count	Number of merged protected boundaries.	int	19	
\$.quality_checks	Recorded quality checks value for this record	object	{...}	
\$.quality_checks.no_empty_candidate_statements	Recorded no empty candidate statements value for this record	bool	True	
\$.quality_checks.candidate_statement_order_is_sequential	Recorded candidate statement order is sequential value for this record	bool	True	
\$.quality_checks.every_statement_has_valid_source_element_id	Identifier of every statement has valid source element.	bool	True	
\$.quality_checks.every_nonempty_processed_element_produces_statement	Recorded every nonempty processed element produces statement value for this record	bool	True	
\$.quality_checks.character_offsets_reproduce_statement_text	Recorded character offsets reproduce statement text value for this record	bool	True	
\$.quality_checks.protected_legal_abbreviation_boundary_violations	Recorded protected legal abbreviation boundary violations value for this record	int	0	
\$.quality_checks.protected_boundary_violation_details	Recorded protected boundary violation details value for this record	array	[...]	
\$.warning_summary	Recorded warning summary value for this record	object	{...}	
\$.warning_summary.missing_terminal_punctuation	Recorded missing terminal punctuation value for this record	int	119	
\$.warning_summary.sentence_starts_lowercase	Recorded sentence starts lowercase value for this record	int	24	
\$.warning_summary.very_short_sentence	Recorded very short sentence value for this record	int	8	
\$.merged_protected_boundaries	Recorded merged protected boundaries value for this record	array	[...]	
\$.merged_protected_boundaries[]	One item in the merged_protected_boundaries array.	object	{...}	

JSON path	Description	Type / format	Example
<code>\$.merged_protected_boundaries[source_element_id]</code>	Identifier of the legal-structure element from which a candidate was segmented.	str	CL_MMA_DEC_000001_PREAMBLE_CONSIDERANDO_P005
<code>\$.merged_protected_boundaries[left_end]</code>	Recorded left end value for this record	int	3
<code>\$.merged_protected_boundaries[right_start]</code>	Recorded right start value for this record	int	3
<code>\$.merged_protected_boundaries[reason]</code>	Recorded reason value for this record	str	nonterminal_or_linebreak
<code>\$.candidate_statements</code>	Recorded candidate statements value for this record	array	[...]
<code>\$.candidate_statements[]</code>	One item in the candidate_statements array.	object	{...}
<code>\$.candidate_statements[statement_id]</code>	Stable identifier of the candidate statement.	str	CL_MMA_DEC_000001_STMT_000001
<code>\$.candidate_statements[segmentation_id]</code>	Identifier of the candidate-statement segmentation event.	str	SEG_000003
<code>\$.candidate_statements[document_id]</code>	Permanent identifier of the source document.	str	CL_MMA_DEC_000001
<code>\$.candidate_statements[structure_id]</code>	Identifier of the legal-structure event.	str	STRUCT_000010
<code>\$.candidate_statements[source_element_id]</code>	Identifier of the legal-structure element from which a candidate was segmented.	str	CL_MMA_DEC_000001_PREAMBLE_VISTO_P001
<code>\$.candidate_statements[source_parent_id]</code>	Identifier of the source element's parent.	str	CL_MMA_DEC_000001_PREAMBLE_VISTO
<code>\$.candidate_statements[source_element_type]</code>	Source element type retained for lineage.	str	paragraph
<code>\$.candidate_statements[source_element_order]</code>	Source element order retained for lineage.	int	12
<code>\$.candidate_statements[sentence_order_within_element]</code>	Sentence position within the source structural element.	int	1
<code>\$.candidate_statements[statement_order]</code>	Global sequential order of the candidate in the document.	int	1
<code>\$.candidate_statements[statement_text]</code>	Original candidate-statement text; never overwritten by transformed or predicted text.	str	Lo dispuesto en la ley N°19.300, sobre Bases Generales del Medio Ambi...
<code>\$.candidate_statements[source_char_start]</code>	Inclusive starting character offset in the source element text.	int	0
<code>\$.candidate_statements[source_char_end]</code>	Exclusive ending character offset in the source element text.	int	2015
<code>\$.candidate_statements[page_start]</code>	Recorded page start value for this record	int	1
<code>\$.candidate_statements[page_end]</code>	Recorded page end value for this record	int	1
<code>\$.candidate_statements[page_assignment_method]</code>	Method used for page assignment.	str	element_inherited

JSON path	Description	Type / format	Example
<code>\$.candidate_statements[].segmentation_method</code>	Method used for segmentation.	str	spacy_sentence_boundaries_with_legal_postprocessing
<code>\$.candidate_statements[].segmentation_status</code>	Completion or review status for segmentation.	str	completed_with_warnings
<code>\$.candidate_statements[].requires_manual_review</code>	Indicates whether human review is required (0 = no; 1 = yes).	int	1
<code>\$.candidate_statements[].review_reason_code</code>	Machine-readable reason for manual review.	str	missing_terminal_punctuation
<code>\$.candidate_statements[].review_reason_note</code>	Human-readable explanation of the review flag.	str	Candidate lacks final period, question mark, or exclamation mark; lis...
<code>\$.candidate_statements[].created_at</code>	Timestamp at which the record was created, in ISO 8601 format.	str	2026-07-23T20:21:02+00:00

2. Candidate-statement CSV

Variable name	Description	Type / format	Example
<code>statement_id</code>	Stable identifier of the candidate statement.	Text	CL_MMA_DEC_000001_STMT_000001
<code>segmentation_id</code>	Identifier of the candidate-statement segmentation event.	Text	SEG_000003
<code>document_id</code>	Permanent identifier of the source document.	Text	CL_MMA_DEC_000001
<code>structure_id</code>	Identifier of the legal-structure event.	Text	STRUCT_000010
<code>source_element_id</code>	Identifier of the legal-structure element from which a candidate was segmented.	Text	CL_MMA_DEC_000001_PREAMBLE_VISTO_P001
<code>source_parent_id</code>	Identifier of the source element's parent.	Text	CL_MMA_DEC_000001_PREAMBLE_VISTO
<code>source_element_type</code>	Source element type retained for lineage.	Text	paragraph
<code>source_element_order</code>	Source element order retained for lineage.	Integer	12
<code>sentence_order_within_element</code>	Sentence position within the source structural element.	Text	1
<code>statement_order</code>	Global sequential order of the candidate in the document.	Integer	1
<code>statement_text</code>	Original candidate-statement text; never overwritten by transformed or predicted text.	Text	Lo dispuesto en la ley N°19.300, sobre Bases Generales del Medio Ambi...
<code>source_char_start</code>	Inclusive starting character offset in the source element text.	Text	0
<code>source_char_end</code>	Exclusive ending character offset in the source element text.	Text	2015
<code>page_start</code>	Recorded page start value for this record	Text	1
<code>page_end</code>	Recorded page end value for this record	Text	1
<code>page_assignment_method</code>	Method used for page assignment.	Text	element_inherited
<code>segmentation_method</code>	Method used for segmentation.	Text	spacy_sentence_boundaries_with_legal_postprocessing
<code>segmentation_status</code>	Completion or review status for segmentation.	Text	completed_with_warnings

Variable name	Description	Type / format	Example
requires_manual_review	Indicates whether human review is required (0 = no; 1 = yes).	Text	1
review_reason_code	Machine-readable reason for manual review.	Text	missing_terminal_punctuation
review_reason_note	Human-readable explanation of the review flag.	Text	Candidate lacks final period, question mark, or exclamation mark; lis...
created_at	Timestamp at which the record was created, in ISO 8601 format.	Text	2026-07-23T20:21:02+00:00

TASK 3: CLASSIFICATION OF CANDIDATE STATEMENTS

Sub-Task 3.1: matching_events

One row per matching and human-review workflow used to construct the labeled corpus.

Implemented foreign-key links: `document_id` → `document_inventory.document_id`; `normalization_id` → `normalization_events.normalization_id`; `structure_id` → `structure_events.structure_id`; `segmentation_id` → `segmentation_events.segmentation_id`.

Variable name	Description	Type / format	Example
matching_id	Identifier of the matching and human-review event.	Text / identifier / primary key	MATCH_000001
segmentation_id	Identifier of the candidate-statement segmentation event. Required in the implemented schema.	Text / identifier	SEG_000003
structure_id	Identifier of the legal-structure event. Required in the implemented schema.	Text / identifier	STRUCT_000010
normalization_id	Identifier of the normalization event. Required in the implemented schema.	Text / identifier	NORM_000004
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
matching_date	Date associated with matching, in ISO format. Required in the implemented schema.	Text / ISO date	2026-08-03
matching_tool	Software or script used for matching. Required in the implemented schema.	Text	ChatGPT-assisted semiautomatic workflow
matching_tool_version	Version identifier for matching tool. Required in the implemented schema.	Text	not recorded
matching_version	Version identifier for matching. Required in the implemented schema.	Text	1.0
reference_file_name	Recorded reference file name value for this one row per matching and human-review workflow used to construct the labeled corpus. Required in the implemented schema.	Text	PLAN_INSUMO.xlsx
reference_file_path	Project-relative path to reference file. Required in the implemented schema.	Text / project-relative path	documents/goal/PLAN_INSUMO.xlsx
reference_drive_id	Google Drive identifier for reference.	Text / identifier	161G8kJ08pJnQLZ9Kgi7-IX7aXFHUrcE
reference_drive_url	URL for reference drive.	Text / URL	https://drive.google.com/file/d/161G8kJ08pJnQLZ9Kgi7-IX7aXFHUrcE
reference_hash_sha256	SHA-256 integrity hash of reference. Required in the implemented schema.	Text / 64-character SHA-256	fd9f8a74973c...

Variable name	Description	Type / format	Example
candidate_csv_path	Project-relative path to candidate csv. Required in the implemented schema.	Text / project-relative path	tables/statement_segmentation/CL_MM_A_DEC_000001/STRUCT_000010_SEG_000...
candidate_drive_id	Google Drive identifier for candidate.	Text / identifier	14uCTUkCtvRmEUvSOFNY8iyXxa57EQHgS
candidate_drive_url	URL for candidate drive.	Text / URL	https://drive.google.com/file/d/14uCTUkCtvRmEUvSOFNY8iyXxa57EQHgS
candidate_hash_sha256	SHA-256 integrity hash of candidate. Required in the implemented schema.	Text / 64-character SHA-256	48c408712cd2...
candidate_statements_count	Number of candidate statements. Required in the implemented schema.	Integer	767
reference_statements_count	Number of reference statements. Required in the implemented schema.	Integer	365
alignment_groups_count	Number of alignment groups. Required in the implemented schema.	Integer	665
review_records_count	Number of review records. Required in the implemented schema.	Integer	768
linked_candidate_records_count	Number of linked candidate records. Required in the implemented schema.	Integer	438
unmatched_candidate_records_count	Number of unmatched candidate records. Required in the implemented schema.	Integer	51
outside_scope_candidate_records_count	Number of outside scope candidate records. Required in the implemented schema.	Integer	278
workbook_only_records_count	Number of workbook only records. Required in the implemented schema.	Integer	1
matching_method	Method used for matching. Required in the implemented schema.	Text	Order-constrained dynamic alignment using normalized textual similari...
normalization_rules_json	JSON object describing the normalization or matching-only transformations applied. Required in the implemented schema.	Text / serialized JSON	{"matching_only":true,"remove_accents":true,"lowercase":true,"standar...
similarity_formula	Recorded similarity formula value for this one row per matching and human-review workflow used to construct the labeled corpus. Required in the implemented schema.	Text	0.34*token_containment + 0.31*token_Dice + 0.28*token_bigram_Dice + 0...
alignment_parameters_json	JSON object describing matching block sizes, thresholds, penalties, and ordering constraints. Required in the implemented schema.	Text / serialized JSON	{"order_constraint":"document_order","candidate_block_max":4,"refere...
label_mapping_json	JSON object mapping reference-workbook labels to modeling labels. Required in the implemented schema.	Text / serialized JSON	{"constitutiva":"constitutive","named_regulative_rule_types":"regulat...
match_type_counts_json	JSON object containing counts by matching relationship type. Required in the implemented schema.	Text / serialized JSON	{"exact":228,"fuzzy":23,"split":154,"merge":23,"complex":10,"unmatche...
review_outcome_counts_json	JSON object summarizing inclusion and reviewer-label outcomes. Required in the implemented schema.	Text / serialized JSON	{"include_in_final_set":{"yes":703,"no":43,"hold":22},"reviewer_gold_...
alignment_review_path	Project-relative path to alignment review. Required in the implemented schema.	Text / project-relative path	outputs/reports/IG_statement_alignment_review.xlsx
alignment_review_hash_sha256	SHA-256 integrity hash of alignment review. Required in the implemented schema.	Text / 64-character SHA-256	7e88d9f58f0c...

Variable name	Description	Type / format	Example
alignment_summary_path	Project-relative path to alignment summary. Required in the implemented schema.	Text / project-relative path	outputs/reports/IG_statement_alignment_summary.xlsx
alignment_summary_hash_sha256	SHA-256 integrity hash of alignment summary. Required in the implemented schema.	Text / 64-character SHA-256	badfde405676...
matching_status	Completion or review status for matching. Required in the implemented schema.	Text	Completed with human review decisions recorded
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-08-04T16:34:04+00:00
notes	Optional free-text warnings, limitations, or implementation notes.	Text	The review workbook was created with ChatGPT; the exact model/tool ve...

Sub-Task 3.2: split_events

One row per reproducible training/test partitioning event.

Implemented foreign-key links: document_id → document_inventory.document_id; segmentation_id → segmentation_events.segmentation_id; matching_id → matching_events.matching_id.

Variable name	Description	Type / format	Example
split_id	Identifier of the dataset-partitioning event.	Text / identifier / primary key	SPLIT_000005
matching_id	Identifier of the matching and human-review event. Required in the implemented schema.	Text / identifier	MATCH_000001
segmentation_id	Identifier of the candidate-statement segmentation event. Required in the implemented schema.	Text / identifier	SEG_000003
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
split_date	Date associated with split, in ISO format. Required in the implemented schema.	Text / ISO date	2026-08-04
split_tool	Software or script used for split. Required in the implemented schema.	Text	07_create_train_test_split.py
split_tool_version	Version identifier for split tool. Required in the implemented schema.	Text	1.3

Variable name	Description	Type / format	Example
source_review_path	Project-relative path to source review. Required in the implemented schema.	Text / project-relative path	outputs/reports/IG_statement_alignment_review.xlsx
source_review_hash_sha256	SHA-256 integrity hash of source review. Required in the implemented schema.	Text / 64-character SHA-256	7e88d9f58f0c...
source_summary_path	Project-relative path to source summary. Required in the implemented schema.	Text / project-relative path	outputs/reports/IG_statement_alignment_summary.xlsx
source_summary_hash_sha256	SHA-256 integrity hash of source summary. Required in the implemented schema.	Text / 64-character SHA-256	badfde405676...
eligibility_rules_json	JSON array or object describing inclusion rules for the modelable corpus. Required in the implemented schema.	Text / serialized JSON	["include_in_final_set normalized to yes","reviewer_gold_label is reg...
valid_labels_json	Serialized JSON value containing valid labels. Required in the implemented schema.	Text / serialized JSON	["regulative","constitutive","non_institutional"]
eligible_records_count	Number of eligible records. Required in the implemented schema.	Integer	702
excluded_records_count	Number of excluded records. Required in the implemented schema.	Integer	66
training_records_count	Number of training records. Required in the implemented schema.	Integer	561
test_records_count	Number of test records. Required in the implemented schema.	Integer	141
class_counts_json	Serialized JSON value containing class counts. Required in the implemented schema.	Text / serialized JSON	{"eligible":{"constitutive":154,"non_institutional":209,"regulative":...

Variable name	Description	Type / format		Example
requested_test_proportion	Recorded requested test proportion value for this one row per reproducible training/test partitioning event. Required in the implemented schema.	Real number	0.2000	
observed_test_proportion	Recorded observed test proportion value for this one row per reproducible training/test partitioning event. Required in the implemented schema.	Real number	0.2009	
random_seed	Fixed random seed used to reproduce the procedure. Required in the implemented schema.	Integer	42	
stratification_field	Recorded stratification field value for this one row per reproducible training/test partitioning event. Required in the implemented schema.	Text	reviewer_gold_label	
grouping_field	Recorded grouping field value for this one row per reproducible training/test partitioning event. Required in the implemented schema.	Text	duplicate_group_id	
split_method	Method used for split. Required in the implemented schema.	Text	deterministic_weighted_duplicate_group_allocation	
duplicate_group_normalization	Recorded duplicate group normalization value for this one row per reproducible training/test partitioning event. Required in the implemented schema.	Text	Unicode NFKC; lowercase; collapse whitespace; trim	
near_duplicate_threshold	Recorded near duplicate threshold value for this one row per reproducible training/test partitioning event. Required in the implemented schema.	Real number	0.9000	

Variable name	Description	Type / format	Example
duplicate_audit_json	JSON object summarizing exact and near-duplicate analysis. Required in the implemented schema.	Text / serialized JSON	{"conflicting_label_groups":0,"cross_label_near_duplicate_text_group_...
quality_checks_json	JSON object containing technical and partition-quality checks. Required in the implemented schema.	Text / serialized JSON	{"checks":{"all_eligible_candidate_ids_assigned_once":true,"candidate...
training_csv_path	Project-relative path to training csv. Required in the implemented schema.	Text / project-relative path	outputs/packages/task_3_statement_classification/history/SPLIT_000005...
training_csv_hash_sha256	SHA-256 integrity hash of training csv. Required in the implemented schema.	Text / 64-character SHA-256	a76513bea648...
test_csv_path	Project-relative path to test csv. Required in the implemented schema.	Text / project-relative path	outputs/packages/task_3_statement_classification/history/SPLIT_000005...
test_csv_hash_sha256	SHA-256 integrity hash of test csv. Required in the implemented schema.	Text / 64-character SHA-256	01f0dce1eb00...
split_workbook_path	Project-relative path to split workbook. Required in the implemented schema.	Text / project-relative path	outputs/packages/task_3_statement_classification/history/SPLIT_000005...
split_workbook_hash_sha256	SHA-256 integrity hash of split workbook. Required in the implemented schema.	Text / 64-character SHA-256	cf850f398c86...
manifest_path	Project-relative path to manifest. Required in the implemented schema.	Text / project-relative path	outputs/packages/task_3_statement_classification/history/SPLIT_000005...
manifest_hash_sha256	SHA-256 integrity hash of manifest. Required in the implemented schema.	Text / 64-character SHA-256	dbbc78c8a6a6...
split_status	Completion or review status for split. Required in the implemented schema.	Text	Completed with warnings

Variable name	Description	Type / format	Example
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-08-04T17:12:19+00:00
notes	Optional free-text warnings, limitations, or implementation notes.	Text	Append-only dataset snapshot with immutable run-specific files. Full ...

Sub-Task 3.2: split_assignments

One row per candidate assigned to training or test in a split event.

Implemented foreign-key links: `segmentation_id` → `candidate_statements.segmentation_id`; `candidate_id` → `candidate_statements.statement_id`; `split_id` → `split_events.split_id`.

Variable name	Description	Type / format	Example
split_id	Identifier of the dataset-partitioning event. Required in the implemented schema.	Text / identifier / primary key	SPLIT_000005
segmentation_id	Identifier of the candidate-statement segmentation event. Required in the implemented schema.	Text / identifier	SEG_000003
review_record_id	Stable identifier of the human-review record used to construct the modeling corpus. Required in the implemented schema.	Text / identifier / composite primary key part 2	REV_0001
candidate_id	Stable Task 2 candidate identifier; corresponds to <code>candidate_statements.statement_id</code> . Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001_STMT_000001
partition	Assigned dataset partition: training or test. Required in the implemented schema.	Text	test; train
gold_label	Human-validated target class used for modeling. Required in the implemented schema.	Text	constitutive; non_institutional; regulative
duplicate_group_id	Identifier used to keep exact or normalized-text duplicates in the same partition. Required in the implemented schema.	Text / identifier	DUP_B1438C2E030F2D6E
duplicate_group_size	Number of reviewed observations in the duplicate group. Required in the implemented schema.	Integer	1
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-08-04T17:12:19+00:00

Sub-Task 3.3: vectorization_events

One row per TF-IDF fitting and transformation run.

Implemented foreign-key links: `document_id` → `document_inventory.document_id`; `segmentation_id` → `segmentation_events.segmentation_id`; `matching_id` → `matching_events.matching_id`; `split_id` → `split_events.split_id`.

Variable name	Description	Type / format	Example
vectorization_id	Identifier of the TF-IDF vectorization event.	Text / identifier / primary key	VECT_000001
split_id	Identifier of the dataset-partitioning event. Required in the implemented schema.	Text / identifier	SPLIT_000005
matching_id	Identifier of the matching and human-review event. Required in the implemented schema.	Text / identifier	MATCH_000001
segmentation_id	Identifier of the candidate-statement segmentation event. Required in the implemented schema.	Text / identifier	SEG_000003
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
vectorization_date	Date associated with vectorization, in ISO format. Required in the implemented schema.	Text / ISO date	2026-08-04
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-08-04T19:04:29+00:00
script_name	Recorded script name value for this one row per tf-idf fitting and transformation run. Required in the implemented schema.	Text	08_vectorize_statements.py
script_version	Version identifier for script. Required in the implemented schema.	Text	1.0
vectorization_tool	Software or script used for vectorization. Required in the implemented schema.	Text	scikit-learn TfidfVectorizer

Variable name	Description	Type / format	Example
vectorization_tool_version	Version identifier for vectorization tool. Required in the implemented schema.	Text	1.9.0
input_training_path	Project-relative path to input training. Required in the implemented schema.	Text / project-relative path	outputs/packages/task_3_statement_classification/history/SPLIT_000005...
input_training_hash_sha256	SHA-256 integrity hash of input training. Required in the implemented schema.	Text / 64-character SHA-256	a76513bea648...
input_test_path	Project-relative path to input test. Required in the implemented schema.	Text / project-relative path	outputs/packages/task_3_statement_classification/history/SPLIT_000005...
input_test_hash_sha256	SHA-256 integrity hash of input test. Required in the implemented schema.	Text / 64-character SHA-256	01f0dce1eb00...
input_training_rows	Number of input training. Required in the implemented schema.	Integer	561
input_test_rows	Number of input test. Required in the implemented schema.	Integer	141
class_distributions_json	JSON object recording class counts for the relevant inputs or partitions. Required in the implemented schema.	Text / serialized JSON	{"eligible":{"constitutive":154,"non_institutional":209,"regulative":...
tfidf_parameters_json	JSON object containing the complete fitted TF-IDF configuration. Required in the implemented schema.	Text / serialized JSON	{"analyzer":"word","dtype":"numpy.float64","lowercase":true,"ngram_ra...
tokenization_method	Method used for tokenization. Required in the implemented schema.	Text	scikit-learn word analyzer with the protocol token_pattern; no stemmi...

Variable name	Description	Type / format	Example
spanish_single_character_tokens_json	Serialized JSON value containing spanish single character tokens. Required in the implemented schema.	Text / serialized JSON	["a","e","o","u","y"]
fit_scope	Recorded fit scope value for this one row per tf-idf fitting and transformation run. Required in the implemented schema.	Text	training candidate_text_original only
vocabulary_size	Recorded vocabulary size value for this one row per tf-idf fitting and transformation run. Required in the implemented schema.	Integer	23464
training_matrix_rows	Number of training matrix. Required in the implemented schema.	Integer	561
training_matrix_columns	Number of training matrix. Required in the implemented schema.	Integer	23464
training_matrix_nonzero	Recorded training matrix nonzero value for this one row per tf-idf fitting and transformation run. Required in the implemented schema.	Integer	51210
test_matrix_rows	Number of test matrix. Required in the implemented schema.	Integer	141
test_matrix_columns	Number of test matrix. Required in the implemented schema.	Integer	23464
test_matrix_nonzero	Recorded test matrix nonzero value for this one row per tf-idf fitting and transformation run. Required in the implemented schema.	Integer	8014

Variable name	Description	Type / format	Example
validation_results_json	JSON object recording technical validation checks and their outcomes. Required in the implemented schema.	Text / serialized JSON	{"classifier_training_performed":false,"database_event_corresponds_to...
output_artifacts_count	Number of output artifacts. Required in the implemented schema.	Integer	6
vectorization_status	Completion or review status for vectorization. Required in the implemented schema.	Text	Completed
notes	Optional free-text warnings, limitations, or implementation notes.	Text	The vectorizer was fitted on training text only. Test text and labels...

Sub-Task 3.3: vectorization_artifacts

One row per external vectorization file registered for an event.

Implemented foreign-key links: `vectorization_id` → `vectorization_events.vectorization_id`.

Variable name	Description	Type / format	Example
vectorization_id	Identifier of the TF-IDF vectorization event. Required in the implemented schema.	Text / identifier / primary key	VECT_000001
artifact_role	Machine-readable role of the external artifact within its event. Required in the implemented schema.	Text / composite primary key part 2	features; fitted_vectorizer; test_index; test_matrix; training_index; training_matrix
file_name	File name of the registered artifact. Required in the implemented schema.	Text	X_train_tfidf.npz
artifact_path	Project-relative path to the registered artifact. Required in the implemented schema.	Text / project-relative path	outputs/task_3_statement_classification/vectorization/VECT_000001/X_t...
artifact_hash_sha256	SHA-256 integrity hash of the artifact. Required in the implemented schema.	Text / 64-character SHA-256	9e8bcd48e52...
size_bytes	Artifact size in bytes. Required in the implemented schema.	Integer	246318

Variable name	Description	Type / format	Example
artifact_description	Human-readable description of the artifact's content and role. Required in the implemented schema.	Text	Sparse TF-IDF matrix for training statements.
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-08-04T19:04:29+00:00

Sub-Task 3.4: model_events

One row per model-training and evaluation run; complex configuration and results are stored in JSON fields.

Implemented foreign-key links: `document_id` → `document_inventory.document_id`; `segmentation_id` → `segmentation_events.segmentation_id`; `matching_id` → `matching_events.matching_id`; `split_id` → `split_events.split_id`; `vectorization_id` → `vectorization_events.vectorization_id`.

Variable name	Description	Type / format	Example
model_id	Identifier of the model-training event.	Text / identifier / primary key	MODEL_000001
vectorization_id	Identifier of the TF-IDF vectorization event. Required in the implemented schema.	Text / identifier	VECT_000001
split_id	Identifier of the dataset-partitioning event. Required in the implemented schema.	Text / identifier	SPLIT_000005
matching_id	Identifier of the matching and human-review event. Required in the implemented schema.	Text / identifier	MATCH_000001
segmentation_id	Identifier of the candidate-statement segmentation event. Required in the implemented schema.	Text / identifier	SEG_000003

Variable name	Description	Type / format	Example
document_id	Permanent identifier of the source document. Required in the implemented schema.	Text / identifier	CL_MMA_DEC_000001
model_date	Date associated with model, in ISO format. Required in the implemented schema.	Text / ISO date	2026-08-04
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-08-04T20:21:01+00:00
model_name	Recorded model name value for this one row per model-training and evaluation run Required in the implemented schema.	Text	Model A: proportional Wroblewska-inspired Random Forest
model_designation	Recorded model designation value for this one row per model-training and evaluation run Required in the implemented schema.	Text	Model A
model_description	Recorded model description value for this one row per model-training and evaluation run Required in the implemented schema.	Text	Three-class Random Forest with mutual-information selection retaining...
script_path	Project-relative path to script. Required in the implemented schema.	Text / project-relative path	codes/task_3_statement_classification/09_train_random_forest_model_a.py
script_version	Version identifier for script. Required in the implemented schema.	Text	1.0

Variable name	Description	Type / format	Example
model_tool	Software or script used for model. Required in the implemented schema.	Text	scikit-learn Pipeline, SelectKBest, and RandomForestClassifier
model_tool_version	Version identifier for model tool. Required in the implemented schema.	Text	1.8.0
input_artifacts_json	JSON object identifying the upstream files used by the model run. Required in the implemented schema.	Text / serialized JSON	[{"artifact_hash_sha256":"e9cbec45448bf15823f88725573a3d0a746082a51fb...}]
input_artifacts_count	Number of input artifacts. Required in the implemented schema.	Integer	6
class_order_json	Serialized JSON value containing class order. Required in the implemented schema.	Text / serialized JSON	["regulative","constitutive","non_institutional"]
class_distributions_json	JSON object recording class counts for the relevant inputs or partitions. Required in the implemented schema.	Text / serialized JSON	{"test":{"constitutive":31,"non_institutional":42,"regulative":68},"t...
training_records_count	Number of training records. Required in the implemented schema.	Integer	561
test_records_count	Number of test records. Required in the implemented schema.	Integer	141
reference_vocabulary_size	Recorded reference vocabulary size value for this one row per model-training and evaluation run Required in the implemented schema.	Integer	3770

Variable name	Description	Type / format		Example
reference_selected_feature_count	Number of reference selected feature. Required in the implemented schema.	Integer	70	
retention_proportion	Recorded retention proportion value for this one row per model-training and evaluation run Required in the implemented schema.	Real number	0.0186	
retention_calculation	Recorded retention calculation value for this one row per model-training and evaluation run Required in the implemented schema.	Text	$\text{round}(23464 * 70 / 3770) = 436$	
current_vocabulary_size	Recorded current vocabulary size value for this one row per model-training and evaluation run Required in the implemented schema.	Integer	23464	
current_selected_feature_count	Number of current selected feature. Required in the implemented schema.	Integer	436	
feature_selection_method	Method used for feature selection. Required in the implemented schema.	Text	SelectKBest with reproducible mutual_info_classif, fitted inside each...	
feature_selection_parameters_json	JSON object containing the feature-selection configuration. Required in the implemented schema.	Text / serialized JSON	{ "k":436,"random_state":42,"score_func":"mutual_info_classif" }	

Variable name	Description	Type / format	Example
cross_validation_design_json	JSON object describing fold construction and internal or nested validation design. Required in the implemented schema.	Text / serialized JSON	{"fold_class_distributions":[{"fold":1,"training":{"constitutive":99,...
random_forest_configurations_json	JSON object describing the evaluated Random Forest configurations. Required in the implemented schema.	Text / serialized JSON	{"A1":{"bootstrap":true,"class_weight":null,"configuration_label":"un...
cross_validation_results_json	JSON object summarizing fold-level and aggregate cross-validation results. Required in the implemented schema.	Text / serialized JSON	{"configurations":{"A1":{"mean_accuracy":0.7628476611883692,"mean_mac...
configuration_selection_rule	Recorded configuration selection rule value for this one row per model-training and evaluation run Required in the implemented schema.	Text	Select the greatest mean five-fold CV macro-F1 at 12 decimal places; ...
selected_configuration	Recorded selected configuration value for this one row per model-training and evaluation run Required in the implemented schema.	Text	A1
final_test_metrics_json	JSON object containing metrics calculated on the reserved test partition. Required in the implemented schema.	Text / serialized JSON	{"class_order":["regulative","constitutive","non_institutional"],"con...

Variable name	Description	Type / format		Example
validation_results_json	JSON object recording technical validation checks and their outcomes. Required in the implemented schema.	Text / serialized JSON		{"cross_validation_fold_validation":{"checks":{"five_folds_materializ...
output_artifacts_count	Number of output artifacts. Required in the implemented schema.	Integer	6	
model_status	Completion or review status for model. Required in the implemented schema.	Text	Completed	
notes	Optional free-text warnings, limitations, or implementation notes.	Text		Proportional Wroblewska-inspired adaptation, not an exact reproductio...

Sub-Task 3.4: model_artifacts

One row per fitted, evaluative, or diagnostic file registered for a model event.

Implemented foreign-key links: model_id → model_events.model_id.

Variable name	Description	Type / format		Example
model_id	Identifier of the model-training event. Required in the implemented schema.	Text / identifier / primary key		MODEL_000001
artifact_role	Machine-readable role of the external artifact within its event. Required in the implemented schema.	Text / composite primary key part 2		confusion_matrix; cross_validation_results; fitted_model; performance_report; selected_features; test_predictions; rfecv_feature_curve; rfecv_feature_plot
file_name	File name of the registered artifact. Required in the implemented schema.	Text		model_a_proportional_rf.joblib
artifact_path	Project-relative path to the registered artifact. Required in the implemented schema.	Text / project-relative path		outputs/task_3_statement_classification/models/MODEL_000001/model_a_p...
artifact_hash_sha256	SHA-256 integrity hash of the artifact. Required in the implemented schema.	Text / 64-character SHA-256		642332413537...

Variable name	Description	Type / format	Example
size_bytes	Artifact size in bytes. Required in the implemented schema.	Integer	542412
artifact_description	Human-readable description of the artifact's content and role. Required in the implemented schema.	Text	Final fitted pipeline containing the 436-feature selector and selecte...
created_at	Timestamp at which the record was created, in ISO 8601 format. Required in the implemented schema.	Text / ISO 8601 datetime	2026-08-04T20:21:01+00:00

Task 3 external datasets and registered artifacts

Sub-task	Artifact	Format	Role
3.1	Alignment review	XLSX	Preserves candidate text, reference-workbook text, provisional alignment, reviewer labels, inclusion decisions, and rationale.
3.1	Alignment summary	XLSX	Summarizes inclusion decisions and validated class counts; includes a source-data sheet.
3.2	Training and test datasets	CSV and XLSX	Versioned, duplicate-group-aware labeled partitions used by vectorization and model evaluation.
3.2	Split manifest	JSON	Reproducible record of source hashes, eligibility rules, split design, counts, duplicate audit, quality checks, and output files.
3.3	TF-IDF matrices	NPZ	Sparse training and test matrices; the test matrix is transformed with the training-fitted vectorizer.
3.3	Row mappings	CSV	Links each matrix row to the candidate identifier, original text, gold label, and partition.
3.3	Feature index	CSV	Ordered n-gram features, lengths, and learned IDF values.
3.3	Fitted vectorizer	JOBLIB	Serialized training-fitted TF-IDF vectorizer.
3.4	Fitted model and selector	JOBLIB	Serialized final estimator or pipeline components for each model event.
3.4	Evaluation outputs	CSV, XLSX, PNG	Test predictions, fold results, selected features, confusion matrices, performance workbooks, feature curves, and plots.

3.2 Split manifest JSON

Records the complete source, eligibility, duplicate-control, partition, count, quality, and file-output metadata for SPLIT_000005.

```
{
  "task_id": "task_3_statement_classification",
  "script_id": "07",
  "script_name": "07_create_train_test_split.py",
```

```

"script_version": "1.3",
"run_id": "SPLIT_000005",
"matching_id": "MATCH_000001",
"split_date": "2026-08-04",
"created_at": "2026-08-04T17:12:19+00:00",
"source": {
  "review_workbook": "{}",
  "summary_workbook": "{}",
  "summary_pair_validation": "{}",
  "count_validation": "{}",
  "eligibility_count_validation": "{}",
  "review_adjustment_validation": "{}",
  "candidate_id_validation": "{}",
  "review_adjustment_note": "REV_0752 was corrected from constitutive to
regulative so its reviewe..."
},
"eligibility_rules": [
  "include_in_final_set normalized to yes"
],
"input_text_field": "candidate_text_original",
"classification_target": "reviewer_gold_label",
"metadata_fields_are_model_features": "False",
"eligibility_policy": {
  "expected_eligible_counts": "{}",
  "note": "REV_0768 is a workbook-only gap rather than a valid candidate
state..."
},
"exclusions": {
  "eligible_records": "702",
  "excluded_records": "66",
  "reason_counts": "{}",
  "reason_combination_counts": "{}"
},
"split": {
  "training_proportion_requested": "0.8000",
  "test_proportion_requested": "0.2000",
  "random_seed": "42",
  "stratification_field": "reviewer_gold_label",
  "grouping_field": "duplicate_group_id",
  "alignment_group_id_used_for_grouping": "False",
  "method": "deterministic_weighted_duplicate_group_allocation",
  "target_test_records": "141",
  "achieved_test_records": "141",
  "class_target_test_records": "{}",
  "class_achieved_test_records": "{}",
  "summed_absolute_class_count_deviation": "0.6000"
},
"counts": {
  "eligible_total": "702",
  "training_total": "561",
  "test_total": "141",
  "eligible_by_class": "{}",
  "training_by_class": "{}",
  "test_by_class": "{}"
},
"duplicate_audit": {
  "normalization": "Unicode NFKC; lowercase; collapse whitespace; trim",
  "total_duplicate_groups_including_singletons": "667",
  "multi_record_duplicate_groups": "28",
  "records_in_multi_record_duplicate_groups": "63",
  "surplus_duplicate_records": "35",
  "largest_duplicate_group": "3",
  "conflicting_label_groups": "0",

```

```

    "near_duplicate_method": "difflib.SequenceMatcher on normalized text",
    "near_duplicate_threshold": "0.9000",
    "near_duplicates_automatically_excluded": "False",
    "near_duplicate_text_group_pairs": "56",
    "cross_label_near_duplicate_text_group_pairs": "3"
  },
  "training_review_record_ids": [
    "REV_0001"
  ],
  "training_candidate_ids": [
    "CL_MMA_DEC_000001_STMT_000001"
  ],
  "test_review_record_ids": [
    "REV_0010"
  ],
  "test_candidate_ids": [
    "CL_MMA_DEC_000001_STMT_000010"
  ],
  "quality_checks": {
    "checks": "{}",
    "class_presence": "{}",
    "requested_test_proportion": "0.2000",
    "observed_test_proportion": "0.2009",
    "test_vs_overall_absolute_class_proportion_difference": "{}",
    "duplicate_groups_crossing_partitions": "[...]",
    "design_constraints": "{}"
  },
  "modeling_steps_performed": [],
  "execution_status": "Completed with warnings",
  "output_files": {
    "IG_statement_training_set.csv": "{}",
    "IG_statement_test_set.csv": "{}",
    "IG_statement_train_test_split.xlsx": "{}"
  }
}

```

JSON path	Description	Type / format	Example
\$.task_id	Identifier of task.	str	task 3 statement classification
\$.script_id	Identifier of script.	str	07
\$.script_name	Recorded script name value for this record	str	07_create_train_test_split.py
\$.script_version	Version identifier for script.	str	1.3
\$.run_id	Identifier of run.	str	SPLIT_000005
\$.matching_id	Identifier of the matching and human-review event.	str	MATCH_000001
\$.split_date	Date associated with split, in ISO format.	str	2026-08-04
\$.created_at	Timestamp at which the record was created, in ISO 8601 format.	str	2026-08-04T17:12:19+00:00
\$.source	Recorded source value for this record	object	{...}
\$.source.review_workbook	Recorded review workbook value for this record	object	{...}
\$.source.review_workbook.filename	Recorded filename value for this record	str	IG_statement_alignment_review.xlsx

JSON path	Description	Type / format	Example
<code>\$.source.review_workbook.path</code>	Recorded path value for this record	str	outputs/reports/IG_statement_alignment_review.xlsx
<code>\$.source.review_workbook.sha256</code>	Recorded sha256 value for this record	str	7e88d9f58f0ce487659f8c9459499f628502a39727d422c7711d962d68e1861c
<code>\$.source.review_workbook.sheet</code>	Recorded sheet value for this record	str	Alignment Review
<code>\$.source.review_workbook.header_row</code>	Recorded header row value for this record	int	5
<code>\$.source.summary_workbook</code>	Recorded summary workbook value for this record	object	{...}
<code>\$.source.summary_workbook.filename</code>	Recorded filename value for this record	str	IG_statement_alignment_summary.xlsx
<code>\$.source.summary_workbook.path</code>	Recorded path value for this record	str	outputs/reports/IG_statement_alignment_summary.xlsx
<code>\$.source.summary_workbook.sha256</code>	Recorded sha256 value for this record	str	badfde4056769faba3a085b36b94a05e74a1ed7336207a7601b2942bbb5c36fc
<code>\$.source.summary_workbook.source_data_sheet</code>	Source data sheet retained for lineage.	str	Source Data
<code>\$.source.summary_pair_validation</code>	Recorded summary pair validation value for this record	object	{...}
<code>\$.source.summary_pair_validation.status</code>	Recorded status value for this record	str	passed
<code>\$.source.summary_pair_validation.compared_records</code>	Number of compared.	int	768
<code>\$.source.summary_pair_validation.compared_columns</code>	Number of compared.	array	[...]
<code>\$.source.summary_pair_validation.compared_columns[]</code>	One item in the compared_columns array.	str	review_record_id
<code>\$.source.summary_pair_validation.differing_cells</code>	Recorded differing cells value for this record	int	0
<code>\$.source.count_validation</code>	Recorded count validation value for this record	object	{...}
<code>\$.source.count_validation.status</code>	Recorded status value for this record	str	overridden
<code>\$.source.count_validation.expected</code>	Recorded expected value for this record	object	{...}
<code>\$.source.count_validation.expected.total_review_records</code>	Number of total review.	int	768
<code>\$.source.count_validation.expected.include_in_final_set</code>	Recorded include in final set value for this record	object	{...}
<code>\$.source.count_validation.expected.included_reviewer_gold_label</code>	Recorded included reviewer gold label value for this record	object	{...}
<code>\$.source.count_validation.observed</code>	Recorded observed value for this record	object	{...}
<code>\$.source.count_validation.observed.total_review_records</code>	Number of total review.	int	768

JSON path	Description	Type / format		Example
<code>\$.source.count_validation.observed.include_in_final_set</code>	Recorded include in final set value for this record	object	{...}	
<code>\$.source.count_validation.observed.included_reviewer_gold_label</code>	Recorded included reviewer gold label value for this record	object	{...}	
<code>\$.source.count_validation.observed.unexpected_include_values</code>	Recorded unexpected include values value for this record	object	{...}	
<code>\$.source.count_validation.observed.unexpected_included_labels</code>	Recorded unexpected included labels value for this record	object	{...}	
<code>\$.source.count_validation.mismatches</code>	Recorded mismatches value for this record	array	[...]	
<code>\$.source.count_validation.mismatches[]</code>	One item in the mismatches array.	str		included_reviewer_gold_label.regulative: expected 338, observed 339
<code>\$.source.eligibility_count_validation</code>	Recorded eligibility count validation value for this record	object	{...}	
<code>\$.source.eligibility_count_validation.status</code>	Recorded status value for this record	str		passed
<code>\$.source.eligibility_count_validation.expected</code>	Recorded expected value for this record	object	{...}	
<code>\$.source.eligibility_count_validation.expected.eligible_records</code>	Number of eligible.	int		702
<code>\$.source.eligibility_count_validation.expected.reviewers_gold_label</code>	Recorded reviewer gold label value for this record	object	{...}	
<code>\$.source.eligibility_count_validation.observed</code>	Recorded observed value for this record	object	{...}	
<code>\$.source.eligibility_count_validation.observed.eligible_records</code>	Number of eligible.	int		702
<code>\$.source.eligibility_count_validation.observed.reviewers_gold_label</code>	Recorded reviewer gold label value for this record	object	{...}	
<code>\$.source.eligibility_count_validation.mismatches</code>	Recorded mismatches value for this record	array	[...]	
<code>\$.source.eligibility_count_validation.policy_note</code>	Recorded policy note value for this record	str		REV_0768 is a workbook-only gap rather than a valid candidate state...
<code>\$.source.review_adjustment_validation</code>	Recorded review adjustment validation value for this record	object	{...}	
<code>\$.source.review_adjustment_validation.status</code>	Recorded status value for this record	str		passed
<code>\$.source.review_adjustment_validation.checks</code>	Recorded checks value for this record	object	{...}	
<code>\$.source.review_adjustment_validation.checks.REV_0283_label_regulative</code>	Recorded REV 0283 label regulative value for this record	bool		True

JSON path	Description	Type / format		Example
<code>\$.source.review_adjustment_validation.checks.REV_0752_label_regulative</code>	Recorded REV 0752 label regulative value for this record	bool	True	
<code>\$.source.review_adjustment_validation.checks.REV_0752_matches_REV_0283_normalized_text</code>	Recorded REV 0752 matches REV 0283 normalized text value for this record	bool	True	
<code>\$.source.review_adjustment_validation.checks.REV_0768_included_yes</code>	Recorded REV 0768 included yes value for this record	bool	True	
<code>\$.source.review_adjustment_validation.checks.REV_0768_blank_candidate_id</code>	Identifier of REV 0768 blank candidate.	bool	True	
<code>\$.source.review_adjustment_validation.checks.REV_0768_blank_candidate_text_original</code>	Recorded REV 0768 blank candidate text original value for this record	bool	True	
<code>\$.source.review_adjustment_validation.source_review_adjustment_note</code>	Source review adjustment note retained for lineage.	str	REV_0752 was corrected from constitutive to regulative so its reviewe...	
<code>\$.source.review_adjustment_validation.eligibility_policy_note</code>	Recorded eligibility policy note value for this record	str	REV_0768 is a workbook-only gap rather than a valid candidate stateme...	
<code>\$.source.candidate_id_validation</code>	Recorded candidate id validation value for this record	object	{...}	
<code>\$.source.candidate_id_validation.source_records</code>	Number of source.	int	768	
<code>\$.source.candidate_id_validation.nonblank_candidate_ids</code>	Recorded nonblank candidate ids value for this record	int	767	
<code>\$.source.candidate_id_validation.unique_nonblank_candidate_ids</code>	Recorded unique nonblank candidate ids value for this record	int	767	
<code>\$.source.candidate_id_validation.blank_candidate_ids</code>	Recorded blank candidate ids value for this record	int	1	
<code>\$.source.review_adjustment_note</code>	Recorded review adjustment note value for this record	str	REV_0752 was corrected from constitutive to regulative so its reviewe...	
<code>\$.eligibility_rules</code>	Recorded eligibility rules value for this record	array	[...]	
<code>\$.eligibility_rules[]</code>	One item in the eligibility_rules array.	str	include_in_final_set normalized to yes	
<code>\$.input_text_field</code>	Recorded input text field.	str	candidate_text_original	
<code>\$.classification_target</code>	Recorded classification target value for this record	str	reviewer_gold_label	

JSON path	Description	Type / format		Example
<code>\$.metadata_fields_are_model_features</code>	Recorded metadata fields are model features value for this record	bool	False	
<code>\$.eligibility_policy</code>	Recorded eligibility policy value for this record	object	{...}	
<code>\$.eligibility_policy.expected_eligible_counts</code>	Recorded expected eligible counts value for this record	object	{...}	
<code>\$.eligibility_policy.expected_eligible_counts.eligible_records</code>	Number of eligible.	int	702	
<code>\$.eligibility_policy.expected_eligible_countsreviewer_gold_label</code>	Recorded reviewer gold label value for this record	object	{...}	
<code>\$.eligibility_policy.expected_eligible_countsreviewer_gold_label.regulative</code>	Recorded regulative value for this record	int	339	
<code>\$.eligibility_policy.expected_eligible_countsreviewer_gold_label.constitutive</code>	Recorded constitutive value for this record	int	154	
<code>\$.eligibility_policy.expected_eligible_countsreviewer_gold_label.non_institutional</code>	Recorded non institutional value for this record	int	209	
<code>\$.eligibility_policy.note</code>	Recorded note value for this record	str	REV_0768 is a workbook-only gap rather than a valid candidate state...	
<code>\$.exclusions</code>	Recorded exclusions value for this record	object	{...}	
<code>\$.exclusions.eligible_records</code>	Number of eligible.	int	702	
<code>\$.exclusions.excluded_records</code>	Number of excluded.	int	66	
<code>\$.exclusions.reason_counts</code>	Recorded reason counts value for this record	object	{...}	
<code>\$.exclusions.reason_counts.blank_candidate_id</code>	Identifier of blank candidate.	int	1	
<code>\$.exclusions.reason_counts.blank_candidate_text_original</code>	Recorded blank candidate text original value for this record	int	1	
<code>\$.exclusions.reason_counts.include_in_final_set_hold</code>	Recorded include in final set hold value for this record	int	22	
<code>\$.exclusions.reason_counts.include_in_final_set_no</code>	Recorded include in final set no value for this record	int	43	
<code>\$.exclusions.reason_counts.invalid_reviewer_gold_label</code>	Recorded invalid reviewer gold label value for this record	int	47	
<code>\$.exclusions.reason_combination_counts</code>	Recorded reason combination counts value for this record	object	{...}	
<code>\$.exclusions.reason_combination_counts.blank_candidate_text_original; blank_candidate_id</code>	Identifier of blank candidate text original; blank candidate.	int	1	

JSON path	Description	Type / format	Example
<code>\$.exclusions.reason_combination_counts.include_in_final_set_hold</code>	Recorded include in final set hold value for this record	int	13
<code>\$.exclusions.reason_combination_counts.include_in_final_set_hold; invalid_reviewer_gold_label</code>	Recorded include in final set hold; invalid reviewer gold label value for this record	int	9
<code>\$.exclusions.reason_combination_counts.include_in_final_set_no</code>	Recorded include in final set no value for this record	int	5
<code>\$.exclusions.reason_combination_counts.include_in_final_set_no; invalid_reviewer_gold_label</code>	Recorded include in final set no; invalid reviewer gold label value for this record	int	38
<code>\$.split</code>	Recorded split value for this record	object	{...}
<code>\$.split.training_proportion_requested</code>	Recorded training proportion requested value for this record	float	0.8000
<code>\$.split.test_proportion_requested</code>	Recorded test proportion requested value for this record	float	0.2000
<code>\$.split.random_seed</code>	Fixed random seed used to reproduce the procedure.	int	42
<code>\$.split.stratification_field</code>	Recorded stratification field value for this record	str	reviewer_gold_label
<code>\$.split.grouping_field</code>	Recorded grouping field value for this record	str	duplicate_group_id
<code>\$.split.alignment_group_id_used_for_grouping</code>	Recorded alignment group id used for grouping value for this record	bool	False
<code>\$.split.method</code>	Recorded method value for this record	str	deterministic_weighted_duplicate_group_allocation
<code>\$.split.target_test_records</code>	Number of target test.	int	141
<code>\$.split.achieved_test_records</code>	Number of achieved test.	int	141
<code>\$.split.class_target_test_records</code>	Number of class target test.	object	{...}
<code>\$.split.class_target_test_records.regulative</code>	Recorded regulative value for this record	float	67.8000
<code>\$.split.class_target_test_records.constitutive</code>	Recorded constitutive value for this record	float	30.8000
<code>\$.split.class_target_test_records.non_institutional</code>	Recorded non institutional value for this record	float	41.8000
<code>\$.split.class_achieved_test_records</code>	Number of class achieved test.	object	{...}
<code>\$.split.class_achieved_test_records.regulative</code>	Recorded regulative value for this record	int	68

JSON path	Description	Type / format		Example
\$.split.class_achieved_test_records.constitutive	Recorded constitutive value for this record	int	31	
\$.split.class_achieved_test_records.non_institutional	Recorded non institutional value for this record	int	42	
\$.split.summed_absolute_class_count_deviation	Recorded summed absolute class count deviation value for this record	float	0.6000	
\$.counts	Recorded counts value for this record	object	{...}	
\$.counts.eligible_total	Recorded eligible total value for this record	int	702	
\$.counts.training_total	Recorded training total value for this record	int	561	
\$.counts.test_total	Recorded test total value for this record	int	141	
\$.counts.eligible_by_class	Recorded eligible by class value for this record	object	{...}	
\$.counts.eligible_by_class.regulative	Recorded regulative value for this record	int	339	
\$.counts.eligible_by_class.constitutive	Recorded constitutive value for this record	int	154	
\$.counts.eligible_by_class.non_institutional	Recorded non institutional value for this record	int	209	
\$.counts.training_by_class	Recorded training by class value for this record	object	{...}	
\$.counts.training_by_class.regulative	Recorded regulative value for this record	int	271	
\$.counts.training_by_class.constitutive	Recorded constitutive value for this record	int	123	
\$.counts.training_by_class.non_institutional	Recorded non institutional value for this record	int	167	
\$.counts.test_by_class	Recorded test by class value for this record	object	{...}	
\$.counts.test_by_class.regulative	Recorded regulative value for this record	int	68	
\$.counts.test_by_class.constitutive	Recorded constitutive value for this record	int	31	
\$.counts.test_by_class.non_institutional	Recorded non institutional value for this record	int	42	
\$.duplicate_audit	Recorded duplicate audit value for this record	object	{...}	
\$.duplicate_audit.normalization	Recorded normalization value for this record	str	Unicode NFKC; lowercase; collapse whitespace; trim	

JSON path	Description	Type / format	Example
<code>\$.duplicate_audit.total_duplicate_groups_including_singletons</code>	Recorded total duplicate groups including singletons value for this record	int	667
<code>\$.duplicate_audit.multi_record_duplicate_groups</code>	Recorded multi record duplicate groups value for this record	int	28
<code>\$.duplicate_audit.records_in_multi_record_duplicate_groups</code>	Recorded records in multi record duplicate groups value for this record	int	63
<code>\$.duplicate_audit.surplus_duplicate_records</code>	Number of surplus duplicate.	int	35
<code>\$.duplicate_audit.largest_duplicate_group</code>	Recorded largest duplicate group value for this record	int	3
<code>\$.duplicate_audit.conflicting_label_groups</code>	Recorded conflicting label groups value for this record	int	0
<code>\$.duplicate_audit.near_duplicate_method</code>	Method used for near duplicate.	str	difflib.SequenceMatcher on normalized text
<code>\$.duplicate_audit.near_duplicate_threshold</code>	Recorded near duplicate threshold value for this record	float	0.9000
<code>\$.duplicate_audit.near_duplicates_automatically_excluded</code>	Recorded near duplicates automatically excluded value for this record	bool	False
<code>\$.duplicate_audit.near_duplicate_text_group_pairs</code>	Recorded near duplicate text group pairs value for this record	int	56
<code>\$.duplicate_audit.cross_label_near_duplicate_text_group_pairs</code>	Recorded cross label near duplicate text group pairs value for this record	int	3
<code>\$.training_review_record_ids</code>	Recorded training review record ids value for this record	array	[...]
<code>\$.training_review_record_ids[]</code>	One item in the training_review_record_ids array.	str	REV_0001
<code>\$.training_candidate_ids</code>	Recorded training candidate ids value for this record	array	[...]
<code>\$.training_candidate_ids[]</code>	One item in the training_candidate_ids array.	str	CL_MMA_DEC_000001_STMT_000001
<code>\$.test_review_record_ids</code>	Recorded test review record ids value for this record	array	[...]
<code>\$.test_review_record_ids[]</code>	One item in the test_review_record_ids array.	str	REV_0010
<code>\$.test_candidate_ids</code>	Recorded test candidate ids value for this record	array	[...]

JSON path	Description	Type / format		Example
\$.test_candidate_ids[]	One item in the test_candidate_ids array.	str		CL_MMA_DEC_000001_STMT_000010
\$.quality_checks	Recorded quality checks value for this record	object		{...}
\$.quality_checks.checks	Recorded checks value for this record	object		{...}
\$.quality_checks.checks.training_plus_test_equals_eligible	Recorded training plus test equals eligible value for this record	bool		True
\$.quality_checks.checks.candidate_ids_do_not_overlap	Recorded candidate ids do not overlap value for this record	bool		True
\$.quality_checks.checks.all_eligible_candidate_ids_assigned_once	Recorded all eligible candidate ids assigned once value for this record	bool		True
\$.quality_checks.checks.duplicate_groups_do_not_cross_partitions	Recorded duplicate groups do not cross partitions value for this record	bool		True
\$.quality_checks.checks.no_blank_candidate_id	Identifier of no blank candidate.	bool		True
\$.quality_checks.checks.no_blank_candidate_text	Recorded no blank candidate text value for this record	bool		True
\$.quality_checks.checks.no_blank_reviewed_label	Recorded no blank reviewed label value for this record	bool		True
\$.quality_checks.checks.only_approved_labels	Recorded only approved labels value for this record	bool		True
\$.quality_checks.checks.every_class_appears_in_both_sets	Recorded every class appears in both sets value for this record	bool		True
\$.quality_checks.checks.candidate_text_original_unchanged	Recorded candidate text original unchanged value for this record	bool		True
\$.quality_checks.class_presence	Recorded class presence value for this record	object		{...}
\$.quality_checks.class_presence.regulative	Recorded regulative value for this record	object		{...}
\$.quality_checks.class_presence.regulative.training	Recorded training value for this record	int		271
\$.quality_checks.class_presence.regulative.test	Recorded test value for this record	int		68
\$.quality_checks.class_presence.constitutive	Recorded constitutive value for this record	object		{...}
\$.quality_checks.class_presence.constitutive.training	Recorded training value for this record	int		123

JSON path	Description	Type / format		Example
<code>\$.quality_checks.class_presence.constitutive.test</code>	Recorded test value for this record	int	31	
<code>\$.quality_checks.class_presence.non_institutional</code>	Recorded non institutional value for this record	object	{...}	
<code>\$.quality_checks.class_presence.non_institutional.training</code>	Recorded training value for this record	int	167	
<code>\$.quality_checks.class_presence.non_institutional.test</code>	Recorded test value for this record	int	42	
<code>\$.quality_checks.requested_test_proportion</code>	Recorded requested test proportion value for this record	float	0.2000	
<code>\$.quality_checks.observed_test_proportion</code>	Recorded observed test proportion value for this record	float	0.2009	
<code>\$.quality_checks.test_vs_overall_absolute_class_proportion_difference</code>	Recorded test vs overall absolute class proportion difference value for this record	object	{...}	
<code>\$.quality_checks.test_vs_overall_absolute_class_proportion_difference.regulative</code>	Recorded regulative value for this record	float	0.0006	
<code>\$.quality_checks.test_vs_overall_absolute_class_proportion_difference.constitutive</code>	Recorded constitutive value for this record	float	0.0005	
<code>\$.quality_checks.test_vs_overall_absolute_class_proportion_difference.non_institutional</code>	Recorded non institutional value for this record	float	0.0002	
<code>\$.quality_checks.duplicate_groups_crossing_partitions</code>	Recorded duplicate groups crossing partitions value for this record	array	[...]	
<code>\$.quality_checks.design_constraints</code>	Recorded design constraints value for this record	object	{...}	
<code>\$.quality_checks.design_constraints.split_grouping_field</code>	Recorded split grouping field value for this record	str	duplicate_group_id	
<code>\$.quality_checks.design_constraints.alignment_group_id_used_for_grouping</code>	Recorded alignment group id used for grouping value for this record	bool	False	
<code>\$.quality_checks.design_constraints.modeling_components_fitted</code>	Recorded modeling components fitted value for this record	array	[...]	
<code>\$.modeling_steps_performed</code>	Recorded modeling steps performed value for this record	array	[...]	
<code>\$.execution_status</code>	Completion or review status for execution.	str	Completed with warnings	
<code>\$.output_files</code>	Recorded output files value for this record	object	{...}	

JSON path	Description	Type / format	Example
<code>\$.output_files.IG_statement_training_set.csv</code>	Recorded IG statement training set.csv value for this record	object	{...}
<code>\$.output_files.IG_statement_training_set.csv.path</code>	Recorded path value for this record	str	outputs/packages/task_3_statement_classification/IG_statement_trainin...
<code>\$.output_files.IG_statement_training_set.csv.sha256</code>	Recorded sha256 value for this record	str	a76513bea648faf80f89c0d35d424f09282db2f80759807c195dbce8cc406621
<code>\$.output_files.IG_statement_training_set.csv.size_bytes</code>	Artifact size in bytes.	int	222513
<code>\$.output_files.IG_statement_training_set.csv.records</code>	Recorded records value for this record	int	561
<code>\$.output_files.IG_statement_test_set.csv</code>	Recorded IG statement test set.csv value for this record	object	{...}
<code>\$.output_files.IG_statement_test_set.csv.path</code>	Recorded path value for this record	str	outputs/packages/task_3_statement_classification/IG_statement_test_se...
<code>\$.output_files.IG_statement_test_set.csv.sha256</code>	Recorded sha256 value for this record	str	01f0dce1eb00ffe5b5637ac4030b07dbe1d474e7e84da99fe28178fb2f11e10b
<code>\$.output_files.IG_statement_test_set.csv.size_bytes</code>	Artifact size in bytes.	int	53216
<code>\$.output_files.IG_statement_test_set.csv.records</code>	Recorded records value for this record	int	141
<code>\$.output_files.IG_statement_train_test_split.xlsx</code>	Recorded IG statement train test split.xlsx value for this record	object	{...}
<code>\$.output_files.IG_statement_train_test_split.xlsx.path</code>	Recorded path value for this record	str	outputs/packages/task_3_statement_classification/IG_statement_train_t...
<code>\$.output_files.IG_statement_train_test_split.xlsx.sha256</code>	Recorded sha256 value for this record	str	cf850f398c8626a81364a339aa7b2440fdca0bb47855bb0452fe21ac2604f2de
<code>\$.output_files.IG_statement_train_test_split.xlsx.size_bytes</code>	Artifact size in bytes.	int	146661
<code>\$.output_files.IG_statement_train_test_split.xlsx.worksheets</code>	Recorded worksheets value for this record	array	[...]
<code>\$.output_files.IG_statement_train_test_split.xlsx.worksheets[]</code>	One item in the worksheets array.	str	Training

3.2 Training-set CSV

Variable name	Description	Type / format	Example
<code>review_record_id</code>	Stable identifier of the human-review record used to construct the modeling corpus.	Text	REV_0001
<code>candidate_id</code>	Stable Task 2 candidate identifier; corresponds to <code>candidate_statements.statement_id</code> .	Text	CL_MMA_DEC_000001_STMT_000001
<code>alignment_group_id</code>	Identifier of alignment group.	Text	ALG_0388
<code>article</code>	Recorded article value for this record	Text	Preamble
<code>scope</code>	Recorded scope value for this record	Text	preamble
<code>source_parent_id</code>	Identifier of the source element's parent.	Text	CL_MMA_DEC_000001_PREAMBLE_VISTO
<code>match_type</code>	Recorded match type value for this record	Text	outside_workbook_scope
<code>candidate_text_original</code>	Recorded candidate text original value for this record	Text	Lo dispuesto en la ley N°19.300, sobre Bases Generales del Medio Ambi...

Variable name	Description	Type / format	Example
reviewer_gold_label	Recorded reviewer gold label value for this record	Text	non_institutional
split	Recorded split value for this record	Text	training
duplicate_group_id	Identifier used to keep exact or normalized-text duplicates in the same partition.	Text	DUP_B1438C2E030F2D6E
random_seed	Fixed random seed used to reproduce the procedure.	Integer	42

3.2 Test-set CSV

Variable name	Description	Type / format	Example
review_record_id	Stable identifier of the human-review record used to construct the modeling corpus.	Text	REV_0010
candidate_id	Stable Task 2 candidate identifier; corresponds to candidate_statements.statement_id.	Text	CL_MMA_DEC_000001_STMT_000010
alignment_group_id	Identifier of alignment group.	Text	ALG_0397
article	Recorded article value for this record	Text	Preamble
scope	Recorded scope value for this record	Text	preamble
source_parent_id	Identifier of the source element's parent.	Text	CL_MMA_DEC_000001_PREAMBLE_CONSIDERANDO
match_type	Recorded match type value for this record	Text	outside_workbook_scope
candidate_text_original	Recorded candidate text original value for this record	Text	Que, de acuerdo con los resultados de la caracterización físico quími...
reviewer_gold_label	Recorded reviewer gold label value for this record	Text	non_institutional
split	Recorded split value for this record	Text	test
duplicate_group_id	Identifier used to keep exact or normalized-text duplicates in the same partition.	Text	DUP_3360337F2A4BDEB0
random_seed	Fixed random seed used to reproduce the procedure.	Integer	42

3.3 Training row-index CSV

Variable name	Description	Type / format	Example
matrix_row_index	Zero-based row position in the associated sparse matrix.	Integer	0
candidate_id	Stable Task 2 candidate identifier; corresponds to candidate_statements.statement_id.	Text	CL_MMA_DEC_000001_STMT_000001
candidate_text_original	Recorded candidate text original value for this record	Text	Lo dispuesto en la ley N°19.300, sobre Bases Generales del Medio Ambi...
reviewer_gold_label	Recorded reviewer gold label value for this record	Text	non_institutional
split	Recorded split value for this record	Text	training

3.3 Test row-index CSV

Variable name	Description	Type / format	Example
matrix_row_index	Zero-based row position in the associated sparse matrix.	Integer	0
candidate_id	Stable Task 2 candidate identifier; corresponds to candidate_statements.statement_id.	Text	CL_MMA_DEC_000001_STMT_000010
candidate_text_original	Recorded candidate text original value for this record	Text	Que, de acuerdo con los resultados de la caracterización físico quími...
reviewer_gold_label	Recorded reviewer gold label value for this record	Text	non_institutional
split	Recorded split value for this record	Text	test

3.3 TF-IDF feature-index CSV

Variable name	Description	Type / format	Example
feature_index	Recorded feature index value for this record	Integer	0
ngram_text	Recorded ngram text value for this record	Text	000

Variable name	Description	Type / format	Example
ngram_length	Recorded ngram length value for this record	Integer	1
idf_value	Recorded idf value value for this record	Real number	6.232889561225582

3.4 Model test-predictions CSV

Variable name	Description	Type / format	Example
configuration_id	Identifier of configuration.	Text	C1
configuration_label	Recorded configuration label value for this record	Text	nested-RFECV unweighted
matrix_row_index	Zero-based row position in the associated sparse matrix.	Integer	0
candidate_id	Stable Task 2 candidate identifier; corresponds to candidate_statements.statement_id.	Text	CL_MMA_DEC_000001_STMT_000010
candidate_text_original	Recorded candidate text original value for this record	Text	Que, de acuerdo con los resultados de la caracterización físico quími...
gold_label	Human-validated target class used for modeling.	Text	non_institutional
predicted_label	Recorded predicted label value for this record	Text	non_institutional
correct	Recorded correct value for this record	Boolean	True

3.4 Cross-validation results CSV

Variable name	Description	Type / format	Example
accuracy	Recorded accuracy value for this record	Real number	0.8053097345132744
class_weight	Recorded class weight value for this record	Text	None
configuration_id	Identifier of configuration.	Text	A1
configuration_label	Recorded configuration label value for this record	Text	unweighted baseline
feature_scope	Recorded feature scope value for this record	Text	436 mutual-information-selected features
fold	Recorded fold value for this record	Integer	1
macro_f1	Recorded macro f1 value for this record	Real number	0.7791713117800074
macro_precision	Recorded macro precision value for this record	Real number	0.8257309941520466
macro_recall	Recorded macro recall value for this record	Real number	0.7621806298276886
model_family	Recorded model family value for this record	Text	A
record_type	Recorded record type value for this record	Text	fold
selected_configuration	Recorded selected configuration value for this record	Boolean	True
source_model_event_id	Identifier of source model event.	Text	MODEL_000002
training_records	Number of training.	Text	448
validation_records	Number of validation.	Text	113
weighted_f1	Recorded weighted f1 value for this record	Real number	0.8013881078521324
rfecv_selected_features	Recorded rfecv selected features value for this record	Text	null

3.4 Selected-features CSV

Variable name	Description	Type / format	Example
importance_rank	Recorded importance rank value for this record	Text	1
model_feature_position	Recorded model feature position value for this record	Text	1532
original_tfidf_feature_index	Recorded original tfidf feature index value for this record	Integer	6860
ngram_text	Recorded ngram text value for this record	Text	deberán
ngram_length	Recorded ngram length value for this record	Integer	1
feature_selection_method	Method used for feature selection.	Text	complete-training RFECV

Variable name	Description	Type / format	Example
rfecv_ranking	Recorded rfecv ranking value for this record	Text	1
final_random_forest_importance	Recorded final random forest importance value for this record	Real number	0.018288736248648546
idf_value	Recorded idf value value for this record	Real number	3.2043674648485996

3.4 Confusion-matrix CSV

Variable name	Description	Type / format	Example
configuration_id	Identifier of configuration.	Text	C1
configuration_label	Recorded configuration label value for this record	Text	nested-RFECV unweighted
actual_label	Recorded actual label value for this record	Text	regulative
predicted_regulative	Recorded predicted regulative value for this record	Text	63
predicted_constitutive	Recorded predicted constitutive value for this record	Text	1
predicted_non_institutional	Recorded predicted non institutional value for this record	Text	4
actual_total	Recorded actual total value for this record	Integer	68

3.1 Alignment-review workbook fields

Variable name	Description	Type / format	Example
review_record_id	Stable identifier of the human-review record used to construct the modeling corpus.	Text	See IG_statement_alignment_review.xlsx
alignment_group_id	Identifier of alignment group.	Text	See IG_statement_alignment_review.xlsx
candidate_id	Stable Task 2 candidate identifier; corresponds to candidate_statements.statement_id.	Text	See IG_statement_alignment_review.xlsx
candidate_order	Recorded candidate order value for this record	Integer	See IG_statement_alignment_review.xlsx
article	Recorded article value for this record	Text	See IG_statement_alignment_review.xlsx
scope	Recorded scope value for this record	Text	See IG_statement_alignment_review.xlsx
source_parent_id	Identifier of the source element's parent.	Text	See IG_statement_alignment_review.xlsx
candidate_text_original	Recorded candidate text original value for this record	Text	See IG_statement_alignment_review.xlsx
workbook_statement_no	Recorded workbook statement no value for this record	Integer	See IG_statement_alignment_review.xlsx
workbook_text_original	Recorded workbook text original value for this record	Text	See IG_statement_alignment_review.xlsx
workbook_type_original	Recorded workbook type original value for this record	Text	See IG_statement_alignment_review.xlsx
match_type	Recorded match type value for this record	Text	See IG_statement_alignment_review.xlsx
similarity_score	Recorded similarity score value for this record	Text	See IG_statement_alignment_review.xlsx
proposed_gold_label	Recorded proposed gold label value for this record	Text	See IG_statement_alignment_review.xlsx
reviewer_gold_label	Recorded reviewer gold label value for this record	Text	See IG_statement_alignment_review.xlsx
review_status	Completion or review status for review.	Text	See IG_statement_alignment_review.xlsx
include_in_final_set	Recorded include in final set value for this record	Text	See IG_statement_alignment_review.xlsx
reviewer_notes	Recorded reviewer notes value for this record	Text	See IG_statement_alignment_review.xlsx
alignment_rationale	Recorded alignment rationale value for this record	Text	See IG_statement_alignment_review.xlsx
candidate_requires_manual_review	Recorded candidate requires manual review value for this record	Text	See IG_statement_alignment_review.xlsx
candidate_review_reason	Recorded candidate review reason value for this record	Text	See IG_statement_alignment_review.xlsx

RELATIONAL LINEAGE SUMMARY

```
document_inventory
├── extraction_events
│   ├── quality_assessment_events — quality_page_flags
│   │   ├── normalization_events — normalization_page_stats
│   │   │   ├── structure_events — structure_elements
│   │   │   │   ├── segmentation_events — candidate_statements
│   │   │   │   │   ├── matching_events
│   │   │   │   │   │   ├── split_events — split_assignments
│   │   │   │   │   │   │   ├── vectorization_events — vectorization_artifacts
│   │   │   │   │   │   │   │   ├── model_events — model_artifacts
```